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Category Theory and Topos Theory

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Introduction

Category Theory is a mathematical theory that may be said to have been formally born in 1945 with the publication of the paper *General Theory of Natural Equivalences* by Samuel Eilenberg and Saunders Mac Lane [EM45]. The paper expresses the general character of the theory: categories are the means for treating the concept of *naturality* that arises in many areas of Mathematics, of which there are many examples but no formal definition. A category makes it possible to focus the study of various mathematical theories not on the objects they describe, but on their interactions, which are called *morphisms* or arrows: the category of groups **Grp** has group homomorphisms as morphisms, and the category of topological spaces **Top** has continuous maps between topological spaces; similarly, the set $\mathbb{N}$ ordered by $|$ has divisibility relations between natural numbers as its morphisms. A transformation between categories is called a *functor*, and consists in transforming morphisms into morphisms of new categories while preserving their compositions, whereas a *natural transformation* is in turn a transformation between functors that makes it possible to pass from the action of one functor on a morphism to that of the second on the same morphism, *simultaneously* for every morphism. These constructions are typical of Algebraic Topology; indeed, it was also the development of this theory in those years that laid the foundations for the birth of categorical language.

The aim of this work is to examine the most important results on categories and transformations, as well as the principal uses of this language. First, the concepts of *universal* property and limits will be analyzed; these involve the decomposition of certain morphisms always through a unique distinguished morphism, called universal, and the concept of an *adjunction* will also be introduced, which will be seen to be a weak way of identifying two categories. Finally, the last chapter will shift the focus to a particular class of categories, *toposes*, which have much more convenient and familiar properties than other categories, because of their resemblance to the category of sets. *Topos Theory* in fact originated with Grothendieck in order to create a sufficiently good setting for treating geometric-topological problems; however, the definition referred to in this work and in most of the relevant literature is due to Lawvere and Tierney, who make it less restrictive than the original one and allow a wide range of uses of this object, typically in Mathematical Logic and Algebraic Geometry.

Chapter 1

Category Theory

1.1 Categories

In their manifesto for Category Theory [EM45] (p.239), Mac Lane and Eilenberg were careful to distance themselves as much as possible from the foundational problems behind the theory, at least at the time of publication of the paper¹, since the purpose of that publication was quite different, namely to define the concept of naturality, as mentioned above. It has moreover become customary in the literature to give a *naïve* definition of category without mentioning, at least not at first, sets or classes, by saying that a category *encodes the following data* before listing its axioms, thereby leaving aside the nature of the mathematical entity in question. The reason behind this usage is that there is no consensus in the mathematical community as to which set-theoretic definition of category is the most advantageous one [Sch24]. In this work it has therefore been agreed to adopt an approach consisting first of an informal presentation of a category, followed by a set-theoretic one that will serve purely as an example of how a formal definition should actually look. The naive definition *encodes* the axioms of Category Theory, making it free of any particular foundational theory while at the same time comprehensible, just as the naive presentation of sets *encodes* the axioms of ZFC.

Definition 1.1.1. A CATEGORY $\mathbf{C}$ is a mathematical structure that encodes the following data:

- (i) two collections, one $\text{obj}\mathbf{C}$ called the collection of *objects* and the other $\text{mor}\mathbf{C}$ called the collection of *morphisms* (or arrows).
- (ii) for every morphism f , a unique object $\text{dom } f$, called the *domain* of f , and a unique object $\text{cod } f$, called the *codomain* of f . If these are denoted respectively by a and b , this condition will be abbreviated by writing $f : a \rightarrow b$ or $a \xrightarrow{f} b$.
- (iii) for every object c , a morphism $c \xrightarrow{\text{id}_c} c$, called the *identity* of c , often also denoted by 1_c .

¹Mac Lane later agreed in [Mac64] with the definition that he subsequently adopted in his classic [Mac71].

(iv) A partial binary operation of composition between morphisms, in the following sense.

Two morphisms $a \xrightarrow{f} x$ and $b \xrightarrow{g} c$ are *consecutive* (or *composable*) if and only if $\text{cod } f = \text{dom } g$, a condition also represented by writing, more briefly, $a \xrightarrow{f} b \xrightarrow{g} c$. The composition operation assigns to a pair (f, g) of morphisms a morphism fg if and only if f and g are consecutive. Composition satisfies the following properties:

- for every morphism $a \xrightarrow{f} b$, one has $\text{id}_a f = f \text{id}_b = f$;
- for every triple of morphisms $a \xrightarrow{f} b \xrightarrow{g} c \xrightarrow{h} d$ one has $(fg)h = f(gh)$; this makes it possible to omit parentheses in the expression of this morphism, which will often be denoted simply by fgh .

Before continuing with a series of examples intended to convey what a category actually is, we note that the identity of an object is unique, and this can be proved easily: if $c \xrightarrow{e} c$ is an identity of an object c of $\mathbf{C}$, then $e = e1_c = 1_c$.

Example 1.1.2. The category **Set** has sets as objects and functions between sets as morphisms. For every object (set) A , its identity morphism id_A is the identity function on that set, and composition is the usual composition of functions, which is easily checked to satisfy the required properties.

Example 1.1.3. In a similar way one considers the categories:

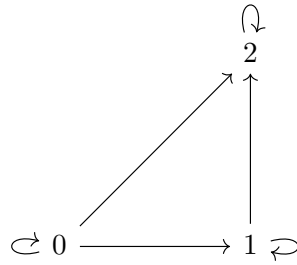
- **Grp**, whose objects are groups and whose morphisms are group homomorphisms;
- **Top**, whose objects are topological spaces and whose morphisms are continuous maps between spaces;
- **Man**, whose objects are smooth manifolds and whose morphisms are smooth maps;
- **Vect $_{\mathbb{K}}$** , whose objects are vector spaces over a fixed field $\mathbb{K}$ and whose morphisms are linear maps.

These categories whose objects are sets equipped with some kind of structure are often called *concrete*. Before proceeding, consider the two topological spaces $(\mathbb{R}, \tau)$ and $(\mathbb{R}, \sigma)$, with τ the Euclidean topology and σ the cofinite topology, and a continuous function f between them. Constructing the category **Top** similarly to the way **Set** is constructed, one might suggest that the domain and codomain of its homomorphisms are those intrinsic to the function, hence in this case the set $\mathbb{R}$, which is the underlying set of both spaces. This leaves an ambiguity in determining domain and codomain; for example, if i is the identity function of $\mathbb{R}$, it is continuous whatever topology is chosen, and therefore would correspond both to $\text{id}_{(\mathbb{R}, \tau)}$ and to $\text{id}_{(\mathbb{R}, \sigma)}$. Thus it should be specified that a morphism in **Top** is a triple that always specifies domain and codomain, namely the morphism should be $((\mathbb{R}, \tau), f, (\mathbb{R}, \sigma))$. In practice, however, this problem is not of great concern in the literature, and so we shall continue to make the usual abuse of terminology and refer to the function, rather than to the triple, as the morphism.

Example 1.1.4. Given a monoid $(M, \cdot)$, one can consider the category $\mathbf{M}$ that has a single object, which will be denoted by X , and whose morphisms are the elements $a \in M$, which will necessarily have the sole object of the category as both domain and codomain: $X \xrightarrow{a} X$. Composition is the operation of the monoid; that is, for any two morphisms $X \xrightarrow{a} X$ and $X \xrightarrow{b} X$, and a third $X \xrightarrow{c} X$, their composition $a \circ b$ will be c if and only if, according to the operation $\cdot$, the corresponding elements satisfy $a \cdot b = c$. The associativity of the operation ensures the associativity of composition between morphisms, and the neutral element $e \in M$ will therefore be the identity morphism $X \xrightarrow{\text{id}_X} X = X \xrightarrow{e} X$. Thus the category $\mathbf{M}$ provides a different perspective from which to view the monoid $(M, \cdot)$.

Example 1.1.5. For every partially ordered set (or *poset*) $(P, \leq)$, one can consider the category $\mathbf{P}$ whose objects are the elements x of P and which has at most one morphism between any pair of objects; more precisely, given two objects x and y , there is a unique arrow $x \longrightarrow y$ if and only if the corresponding elements of P satisfy $x \leq y$. Notice that, since the arrow is unique, it is uniquely determined by its domain and codomain and therefore needs no name. The transitivity of $\leq$ ensures that composition between morphisms is well defined, in the only possible way; reflexivity ensures the existence of identity morphisms.

For example, consider the ordinal $\bar{3} = (\{0, 1, 2\}, \leq)$ as the category $\mathbf{3}$; it can be represented as:



Consider instead the ordinal of all natural numbers with the usual order, $\bar{\omega}$, as a category ω ; it can be represented as follows, omitting compositions and (as is usually done) identities:

$$0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow \cdots \longrightarrow n \longrightarrow \cdots$$

Example 1.1.6. Given a field $\mathbb{K}$, $\mathbf{Mat}_{\mathbb{K}}$ is the category whose morphisms are matrices A over $\mathbb{K}$ and whose objects are the natural numbers, in such a way that the domain of each matrix is its number of rows n and its codomain is its number of columns m . Composition of morphisms is the usual product of matrices:

$$\begin{array}{c} n \longrightarrow m \qquad m \longrightarrow k \qquad n \longrightarrow k \\ \boxed{A} \quad \times \quad \boxed{B} = \boxed{AB} \end{array}$$

The fact that the product satisfies the axioms for composition is fairly trivial to show, but it can be inferred from the well-known fact that matrices represent homomorphisms between

finite-dimensional vector spaces, which form the category $\mathbf{Vect}_{\mathbb{K}}^{\text{fin}}$ (with finite-dimensional vector spaces as objects, linear maps as morphisms, and the usual composition).

Definition 1.1.7. Given a category $\mathbf{C}$ and two of its objects a and b , the HOM-SET of the pair (a, b) is the collection of morphisms of $\mathbf{C}$ of the form $a \xrightarrow{f} b$, which will be denoted by $\mathbf{C}(a, b)$.

Often the categories with which we shall deal below will have as objects sets, or structures having sets as underlying sets, and as morphisms functions that satisfy certain conditions, so in this case it is correct to say that the collection of morphisms between two objects is a set: hom-sets inherit their name for this reason. However, hom-sets are not necessarily sets. For example, if $\mathbf{C}$ is a category having a single object x and as morphisms the proper class of all sets, with composition given by union of sets and identity given by the empty set, then $\mathbf{C}(x, x)$ is a proper class of morphisms.

Definition 1.1.8. A SUBCATEGORY $\mathbf{D}$ of a category $\mathbf{C}$ is a category such that:

- (i) the objects of $\mathbf{D}$ are objects of $\mathbf{C}$ and have the same identities that they have in $\mathbf{C}$;
- (ii) the morphisms of $\mathbf{D}$ are morphisms of $\mathbf{C}$ and have the same domain and codomain that they have in $\mathbf{C}$;
- (iii) $\text{mor}\mathbf{C}$ is closed under composition.

A subcategory $\mathbf{D}$ of a category $\mathbf{C}$ is said to be *full* if any pair of objects a and b of $\mathbf{D}$ has the same morphisms as it has in $\mathbf{C}$, that is, if $\mathbf{D}(a, b) = \mathbf{C}(a, b)$.

Example 1.1.9. Let $\mathbf{CRing}$ be the category of unital commutative rings with morphisms given by unital ring homomorphisms; this is a full subcategory of $\mathbf{Ring}$, the category of unital rings with morphisms given by unital ring homomorphisms. If one also considers the category $\mathbf{Rng}$ of rings, possibly non-unital, with morphisms given by ring homomorphisms, again possibly non-unital, then $\mathbf{Ring}$ is not a full subcategory. In the first case the subcategory is full because the morphisms between unital commutative rings are unital ring homomorphisms between unital commutative rings; in the second case, instead, an arbitrary ring homomorphism, that is, one not necessarily unital, is not necessarily a unital ring homomorphism, since the latter have the additional feature of having to preserve the unit.

The difficulty in giving a unique set-theoretic foundation to Category Theory arises from issues concerning the size of the collections of objects and morphisms under consideration: it is well known that the collection $\text{obj}\mathbf{Set}$ is not a set. Recognizing that some definitions are better than others for the treatment of specific results², we now fix a set-theoretic definition of category.

The concept of *class* in ZFC set theory is not an entity in its own right; it is not, in fact, provided for by the axioms. What one does is instead to identify this kind of

²This issue is discussed thoroughly in [Sch24].

meta-collection with the property that determines all of its elements. Semantically, the property behaves like a genuine collection, and correspondences between classes can be treated as particular predicates. Ultimately, one can practically operate with classes as one does with sets. However, since it is not an object axiomatized by ZFC, a class cannot fall under the scope of any quantifier: one would therefore be unable to make a large number of statements. There are various theories that make it possible to discuss collections with increasing levels of complexity: those at the level immediately above sets are proper classes. We note that an important rule is that only sets can be elements of other sets or classes; therefore, for example, there are no ordered pairs whose coordinates are classes. One way to treat categories using these kinds of *ultra-set-theoretic* devices is to define categories similarly to the naive way, but specifying the nature of the collections of objects and morphisms: categories whose collections of morphisms and objects are sets, categories in which they are proper classes, categories in which the collections are n levels of complexity above sets. In this way one defines categories of ever higher complexity. However, this approach does not allow one really to consider all categories, because of the hierarchy imposed on the collections under consideration: how could one define the category of all categories?

Category Theory could also be treated directly in an axiomatic way, and this approach would be adopted in this work if it did not require a long list of axioms³. Thus most of the results will not be limited by the particular model of the particular set theory chosen, because this determines only the level of complexity and the management of the collections in question. For this reason, the results will be presented within the size bounds of categories with a single level of complexity (classes as collections of objects and morphisms), with the intention of viewing them as an illustrative version of the more general result.

Definition 1.1.10. A DIRECTED GRAPH consists of two classes, one O of elements called *objects* and one A of *arrows*, equipped with two functions between classes $A \xrightarrow{\text{dom}} O$ and $A \xrightarrow{\text{cod}} O$.

Definition 1.1.11. The CLASS OF COMPOSABLE ARROWS of a directed graph with objects in O and arrows in A is the class:

$$A \times_O A = \{(f, g) \in A \times A \mid \text{cod } f = \text{dom } g\}$$

Definition 1.1.12. A CATEGORY $\mathbf{C}$ is a directed graph with objects in O and arrows in A , equipped with two further functions called *identity*

$$\text{id}_- : c \in O \mapsto \text{id}_c \in A$$

and *composition*

$$- \circ - : (f, g) \in A \times_O A \mapsto f \circ g \in A$$

³Lawvere was among the first to speak of this approach to categories in [Law66], thinking of the theory as a way to refund Mathematics.

such that:

- (i) $\text{dom id}_a = \text{cod id}_a = a$, for every object a ;
- (ii) $\text{dom } f \circ g = \text{dom } f$ and $\text{cod } f \circ g = \text{cod } g$, for every pair of composable arrows f and g ;
- (iii) if $\text{dom } f = a$ and $\text{cod } f = b$, then $\text{id}_a \circ f = f \circ \text{id}_b = f$;
- (iv) if f and g are composable, and so are g and h , then $(f \circ g) \circ h = f \circ (g \circ h)$.

A category $\mathbf{C}$ is said to be *small* if $\text{mor}\mathbf{C}$ is a set, and *large* if $\text{mor}\mathbf{C}$ is a proper class. A more common notion is that of a *locally small* category, namely one in which all hom-sets are sets.

1.2 Morphisms

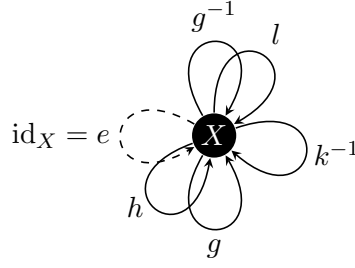
Definition 1.2.1. Given a category $\mathbf{C}$, an ISOMORPHISM is a morphism $a \xrightarrow{f} b$ for which there exists a morphism $b \xrightarrow{g} a$, called the *inverse* of f , such that $fg = \text{id}_a$ and $gf = \text{id}_b$.

It is easy to verify that an inverse morphism is unique: if g and h are inverses of f , then $g = gfh = h$. The inverse morphism of f will be denoted by $b \xrightarrow{f^{-1}} a$. When there exists an isomorphism $a \xrightarrow{f} b$ between two objects a and b of the category $\mathbf{C}$, the two are said to be *isomorphic* in $\mathbf{C}$, and one writes $a \cong b$. We add that a morphism whose domain and codomain are equal, $a \xrightarrow{f} a$, is called an *endomorphism*, and an isomorphism that is an endomorphism is called an *automorphism*.

Example 1.2.2. In $\mathbf{Mat}_{\mathbb{K}}$ an isomorphism is an invertible matrix. An invertible matrix is moreover a square matrix, and hence the isomorphisms all have the same domain and codomain; that is, all isomorphisms are automorphisms.

Example 1.2.3. In $\mathbf{Set}$ the isomorphisms are the bijective functions; in $\mathbf{Grp}$ they are bijective group homomorphisms; in $\mathbf{Top}$ they are homeomorphisms; and in $\mathbf{Man}$ they are smooth diffeomorphisms. Along the same lines, given a first-order language $\mathcal{L}$, consider the category $\mathbf{Str}_{\mathcal{L}}$ whose objects are $\mathcal{L}$ -structures (defined set-theoretically) and whose morphisms are elementary embeddings between structures, with the usual law of composition: the isomorphisms are bijective elementary embeddings, and by choosing the appropriate language one recovers the isomorphisms in the preceding categories. For example, choosing the language $\mathcal{L} = \{\cdot, {}^{-1}, e\}$, two structures G and H that are isomorphic in $\mathbf{Str}_{\mathcal{L}}$ are not necessarily models of group theory, but if they were, they would be isomorphic in the usual sense of groups.

Example 1.2.4. Given a group $(G, \cdot)$, one can define the category $\mathbf{G}$ as was done in Example 1.1.4: the category has a single object X (a one-object category), its morphisms are the elements $g \in G$ (the identity e continues to be the identity morphism), and furthermore they must all be isomorphisms in order to respect the invertibility required by the group axioms.



A category is called a *groupoid* when all of its morphisms are isomorphisms: a one-object group category is an example, and indeed the prototype, of a groupoid. A category whose only morphisms are identities is instead called *discrete*; it too is a groupoid. When, instead, the only isomorphisms in a category are the identities, that is, all objects that are isomorphic to one another are in fact the same object, the category is called *skeletal*: a poset is a skeletal category.

Proposition 1.2.5. *Every category admits a largest groupoid with respect to inclusion of objects and morphisms.*

One can check that this is the subcategory having all the same objects and, as morphisms, only the isomorphisms of the original category.

Example 1.2.6. Given a topological space X , an object of **Top**, let $\Pi_1(X)$ denote the category whose objects are all points of X and whose morphisms are homotopy classes of paths between two points; the composition is concatenation of paths.

Definition 1.2.7. A morphism $a \xrightarrow{f} b$ of a given category **C** is called a **MONOMORPHISM** (or *mono*) when, for any two *parallel* morphisms (with the same domain and codomain) $g, h : x \rightarrow a$, one has:

$$gf = hf \implies g = h$$

whereas f is called an **EPIMORPHISM** (or *epi*) when, for any two parallel morphisms $g, h : b \rightarrow y$, one has:

$$fg = fh \implies g = h$$

If f is *mono* we shall use the notation $a \xrightarrow{f} b$; if it is *epi* we shall write $a \xrightarrow{f} b$.

Example 1.2.8. In **Set** the monomorphisms are exactly the injective functions and the epimorphisms exactly the surjective functions; indeed, the definition of mono and epi in this category is a well-known characterization of injectivity and surjectivity. Observe, for example, that in **Set** every monomorphism $a \xrightarrow{f} b$ is injective, that is, any two elements of a having the same image under f coincide: we shall often use the symbol $* = \{p\}$ to denote an arbitrary singleton, and consider the parallel arrows $x, y : * \rightarrow a$ that send the unique element of $*$ respectively to $x, y \in a$; by what has been said, this diagram

$$\begin{array}{ccc} * & \xrightarrow{x} & a \\ y \downarrow & & \downarrow f \\ a & \xrightarrow{f} & b \end{array}$$

commutes, that is, all compositions of arrows in the diagram having the same domain and codomain coincide: $xf = yf$, whence $x = y$. Commutative diagrams are a typical tool of categorical language and often summarize genuine proofs; indeed, they will be more useful below.

Proposition 1.2.9. *Every isomorphism is both mono and epi.*

Proof. If $a \xrightarrow{f} b$ is an isomorphism, then for two parallel morphisms $g, h : x \rightarrow a$ one has:

$$gf = hf \implies g = g \operatorname{id}_a = gff^{-1} = hff^{-1} = h \operatorname{id}_a = h$$

that is, f is mono. Similarly, for $g, h : b \rightarrow y$ one has:

$$fg = fh \implies g = \operatorname{id}_b g = f^{-1}fg = f^{-1}fh = \operatorname{id}_b h = h$$

that is, f is epi. □

It is well known that in **Set** this condition is also sufficient, yet this is not so in all categories.

Example 1.2.10. In the category **Ring**, the embedding $\mathbb{Z} \xrightarrow{i} \mathbb{Q}$ is both mono and epi. It is a monomorphism because, if we take two parallel morphisms $f, g : R \rightarrow \mathbb{Z}$ such that $fi = gi$, then since i sends all elements of $\mathbb{Z}$ to themselves in $\mathbb{Q}$, it follows trivially that $f = g$. It is an epimorphism because, if we take $f, g : \mathbb{Q} \rightarrow R$ such that $if = ig$, this is equivalent to saying that these homomorphisms restricted to $\mathbb{Z}$ coincide, which by $\mathbb{Z}$ -linearity induces the same behavior on $\mathbb{Q}$ for both f and g . Yet i is not an isomorphism, because there are no nonzero unital morphisms from $\mathbb{Q}$ to $\mathbb{Z}$, and hence no inverse of i exists. From this observation it also follows that, although the objects and morphisms carry an underlying set-theoretic structure, epimorphisms are not necessarily surjective, as in this case.

Example 1.2.11. The category **Field** has fields as objects and field homomorphisms as morphisms, with the usual composition. In **Field** every morphism is a monomorphism. Indeed, a field has only trivial ideals and if $\ker f$ is the kernel of an arbitrary homomorphism $K \xrightarrow{f} F$, it must therefore be the zero ideal (since otherwise one would have the zero map, which, not preserving the unit, is not a homomorphism between fields), that is, f is an injective homomorphism, from which it follows easily that f is mono.

1.3 Duality

Consider specifically the *Elementary Theory of an Abstract Category* (ETAC), which is a first-order theory: its language has no constants, has id_- as its only unary function, and as predicates, besides equality, $\operatorname{mor}-$, $\operatorname{obj}-$, $\operatorname{dom}- = -$, $\operatorname{cod}- = -$ and $- \circ - = -$, while its axioms are those introduced above for an arbitrary category.

Definition 1.3.1. Given a well-formed formula α of ETAC, its DUAL formula is the formula α^* obtained by replacing every occurrence of an atomic formula of the form $\text{dom } f = x$, $\text{cod } f = y$, and $f \circ g = h$ respectively by $\text{cod } f = x$, $\text{dom } f = y$, and $g \circ f = h$.

What is happening with the abbreviations introduced by arrows is essentially their *reversal*. For example, consider the formula

$$\text{mono}(f) = a \xrightarrow{f} b \wedge \forall g \forall h \forall c ((c \xrightarrow{g} a \wedge c \xrightarrow{h} a \wedge g \circ f = h \circ f) \rightarrow (g = h))$$

which obviously expresses the fact that f is a monomorphism; its dual formula will be:

$$\text{mono}(f)^* = b \xrightarrow{f} a \wedge \forall g \forall h \forall c ((a \xrightarrow{g} c \wedge a \xrightarrow{h} c \wedge f \circ g = f \circ h) \rightarrow (g = h))$$

which evidently expresses that f is an epimorphism. In this case one says that by *dualizing* the definition of monomorphism one obtains the definition of epimorphism. Notice also that the dual of the dual of a formula is equal to the original formula, $\alpha^{**} = \alpha$.

Theorem 1.3.2 (Principle of Duality). *If a formula of ETAC is a theorem, then so is its dual formula.*

The proof of the theorem can be carried out by induction on the complexity of the formula, and shows that a proof of the new formula is the sequence of the duals of the formulas in the original proof, preserving their order and the use of the rules of inference. This makes it possible to avoid carrying out a proof of a theorem when one has a proof of the corresponding dual.

This theorem and these considerations can be viewed more broadly, also between multiple categories and through functors: although intuitively one might think of justifying this observation by moving to the (abstract) category of all categories, in fact one needs a stronger theory⁴ The *theory of duality* is a much deeper field of study in categorical logic and goes beyond our purposes; we therefore refer to [Hat82]. In the spirit of this theory, then, we shall introduce new categorical concepts.

Definition 1.3.3. Given a category $\mathbf{C}$, its OPPOSITE CATEGORY is the category $\mathbf{C}^{\text{op}}$ which has the same objects and morphisms as $\mathbf{C}$, with the functions dom and cod interchanged and the partial composition operation replaced by the opposite operation.

More explicitly, $\text{mor } \mathbf{C}$ and $\text{mor } \mathbf{C}^{\text{op}}$ are therefore equal, but each morphism f of $\mathbf{C}$ (and $\mathbf{C}^{\text{op}}$) will be denoted f^{op} when regarded as a morphism of $\mathbf{C}^{\text{op}}$. Given a morphism $a \xrightarrow{f} b$ of $\mathbf{C}$, we have $\text{dom } f = a$ and $\text{cod } f = b$, and hence in $\mathbf{C}^{\text{op}}$ one has $\text{dom } f^{\text{op}} = b$ and $\text{cod } f^{\text{op}} = a$. On the other hand, for every pair of composable morphisms f^{op} and g^{op} , one has $g^{\text{op}} f^{\text{op}} = (fg)^{\text{op}}$. Notice that $\mathbf{C} = (\mathbf{C}^{\text{op}})^{\text{op}}$.

Example 1.3.4. The category ω has ω^{op} as its opposite, and this can be represented as:

$$\cdots \longrightarrow n \longrightarrow \cdots \longrightarrow 3 \longrightarrow 2 \longrightarrow 1 \longrightarrow 0$$

⁴cf. §8 of [Hat82].

In general, the opposite categories of poset categories are the categories obtained from the corresponding posets with the order relation reversed.

Example 1.3.5. The group category $\mathbf{G}$ obtained from $(G, \cdot)$ has as its opposite the category $\mathbf{G}^{\text{op}}$, which has the same single object and morphisms $X \xleftarrow{g^{\text{op}}} X$ for every morphism $X \xrightarrow{g} X$. What actually shows that it can be a category different from the original one is composition: given two objects g and h of $\mathbf{G}$, $g^{\text{op}}h^{\text{op}} = (hg)^{\text{op}}$, where hg corresponds to the element $h \cdot g$ of G . Let $G^{\text{op}} = (G, \cdot^{\text{op}})$ be the *opposite group* of G , with underlying set G and operation defined by: for every $g, h \in G$,

$$g \cdot^{\text{op}} h = h \cdot g$$

It follows easily, therefore, that the opposite category $\mathbf{G}^{\text{op}}$ is in fact the group category obtained from G^{op} : in the case of abelian groups the categories will coincide.

1.4 Functors

Definition 1.4.1. Given two categories $\mathbf{C}$ and $\mathbf{D}$, a COVARIANT FUNCTOR $F: \mathbf{C} \rightarrow \mathbf{D}$ encodes two functions, one from $\text{obj}\mathbf{C}$ to $\text{obj}\mathbf{D}$ and one from $\text{mor}\mathbf{C}$ to $\text{mor}\mathbf{D}$, both denoted by $-^F$, such that:

- (i) the image of every arrow of the form $a \xrightarrow{f} b$ in $\mathbf{C}$ is an arrow of the form $a^F \xrightarrow{f^F} b^F$ in $\mathbf{D}$;
- (ii) for every object c in $\mathbf{C}$, $(\text{id}_c)^F = \text{id}_{c^F}$;
- (iii) for any two composable arrows f and g , $(fg)^F = f^F g^F$.

Given a category $\mathbf{C}$, a functor $F: \mathbf{C} \rightarrow \mathbf{C}$ is called an *endofunctor*. The endofunctor $1_{\mathbf{C}} = \text{id}_{\mathbf{C}}: \mathbf{C} \rightarrow \mathbf{C}$ that leaves every object and morphism unchanged is called the *identity* functor of $\mathbf{C}$.

$$\begin{array}{ccc} \mathbf{C} & & \mathbf{C} \\ \begin{array}{c} a \\ f \downarrow \\ b \end{array} & \xrightarrow{\text{id}_{\mathbf{C}}} & \begin{array}{c} a \\ \downarrow f \\ b \end{array} \end{array}$$

Example 1.4.2. Let $\mathbf{Euclid}_*$ denote the category whose objects are pointed Euclidean spaces $\mathbb{R}_{x_0}^n$ (x_0 will be called the initial point) and whose morphisms are functions differentiable at the initial points and preserving those points, with the usual composition. The functor $\mathbf{Euclid}_* \xrightarrow{D} \mathbf{Mat}_{\mathbb{R}}$ sends every space to its dimension and every function between spaces to the value of its Jacobian matrix at the initial point.

$$\begin{array}{ccc}
\mathbf{Euclid}_* & & \mathbf{Mat}_{\mathbb{R}} \\
\mathbb{R}_{x_0}^n & \xrightarrow{D} & n \\
f \downarrow & & \downarrow Df(x_0) \\
\mathbb{R}_{y_0}^m & & m
\end{array}$$

Example 1.4.3. The fundamental group functor $\mathbf{Top}_* \xrightarrow{\pi_1(-)} \mathbf{Grp}$ assigns to each pointed space X_{x_0} its fundamental group and to each map $X_{x_0} \xrightarrow{f} Y_{y_0}$ the homomorphism $\pi_1(X_{x_0}) \xrightarrow{f^*} \pi_1(Y_{y_0})$, which assigns to each homotopy class $\bar{\gamma}$ of loops in X based at x_0 the class $\overline{\gamma f}$ of loops in Y based at y_0 .

$$\begin{array}{ccc}
\mathbf{Top}_* & & \mathbf{Grp} \\
X_{x_0} & \xrightarrow{\pi_1(-)} & \pi_1(X_{x_0}) \\
f \downarrow & & \downarrow f^* \\
Y_{y_0} & & \pi_1(Y_{y_0})
\end{array}$$

The concept of functoriality comes precisely from Algebraic Topology, since it has to do with the way in which one passes from topological spaces to their algebraic invariants. Another example is the functor $\mathbf{Top} \xrightarrow{H_n(-)} \mathbf{Grp}$, the n -th homology group functor, which assigns to each topological space its homology group of degree n and to each continuous map between two spaces the induced homomorphism between homology groups.

Proposition 1.4.4. *Functors preserve isomorphisms.*

Proof. Let $F: \mathbf{C} \rightarrow \mathbf{D}$ be a functor and $a \xrightarrow{f} b$ an isomorphism in $\mathbf{C}$. Then f^F is an isomorphism because it admits $(f^{-1})^F$ as inverse morphism:

$$\begin{aligned}
\mathrm{id}_a = f f^{-1} &\implies \mathrm{id}_{a^F} = \mathrm{id}_a^F = (f f^{-1})^F = f^F (f^{-1})^F \\
\mathrm{id}_b = f^{-1} f &\implies \mathrm{id}_{b^F} = \mathrm{id}_b^F = (f^{-1} f)^F = (f^{-1})^F f^F
\end{aligned}$$

□

Example 1.4.5. Given the group category $\mathbf{G}$ and an object c of a category $\mathbf{C}$, the functor $\mathbf{G} \xrightarrow{c} \mathbf{C}$ assigns the single object of $\mathbf{G}$ to c and each of its automorphisms to some endomorphism of c , which, by what has been seen, will in turn be an automorphism. This functor essentially encodes the concept of group actions arising in various areas of mathematics, and hence for each different choice of $\mathbf{C}$. If, for example, $\mathbf{C} = \mathbf{Vect}_{\mathbb{K}}$, then by suitably choosing the vector space V , $\mathbf{G} \xrightarrow{V} \mathbf{Vect}_{\mathbb{K}}$, defined as above, is a *linear representation* of the group G .

Example 1.4.6. Given two functors $\mathbf{A} \xrightarrow{F} \mathbf{C}$ and $\mathbf{B} \xrightarrow{G} \mathbf{C}$, the *comma category* $F \downarrow G$ is the category whose objects are triples $(a, f, b) =: f_{a,b}$ as the objects a of $\mathbf{A}$ and b of $\mathbf{B}$ and the arrows $a^F \xrightarrow{f} b^G$ in $\mathbf{C}$ vary, and whose morphisms are pairs $f_{a,b} \xrightarrow{(h,k)} g_{a',b'}$ with $a \xrightarrow{h} a'$ an arrow in $\mathbf{A}$ and $b \xrightarrow{k} b'$ an arrow in $\mathbf{B}$ such that the following diagram commutes:

$$\begin{array}{ccc}
a^F & \xrightarrow{f} & b^G \\
\downarrow h^F & & \downarrow k^G \\
a'^F & \xrightarrow{g} & b'^G
\end{array}$$

Finally, composition is defined componentwise; that is, given morphisms $f_{a,b} \xrightarrow{(h,k)} g_{a',b'} \xrightarrow{(\bar{h},\bar{k})} e_{a'',b''}$, the composite morphism is

$$f_{a,b} \xrightarrow{(h\bar{h},k\bar{k})} e_{a'',b''}$$

This is indeed a morphism in $F \downarrow G$, because since the diagrams

$$\begin{array}{ccc}
a^F & \xrightarrow{f} & b^G \\
\downarrow h^F & & \downarrow k^G \\
a'^F & \xrightarrow{g} & b'^G
\end{array}
\quad
\begin{array}{ccc}
a'^F & \xrightarrow{g} & b'^G \\
\downarrow \bar{h}^F & & \downarrow \bar{k}^G \\
a''^F & \xrightarrow{e} & b''^G
\end{array}$$

commute, the composite morphisms $h^F \bar{h}^F = (h\bar{h})^F$ and $k^G \bar{k}^G = (k\bar{k})^G$ make the diagram

$$\begin{array}{ccc}
a^F & \xrightarrow{f} & b^G \\
\downarrow (h\bar{h})^F & & \downarrow (k\bar{k})^G \\
a''^F & \xrightarrow{e} & b''^G
\end{array}$$

commute; therefore the composite morphism is well defined.

Definition 1.4.7. Given two categories $\mathbf{C}$ and $\mathbf{D}$, a CONTRAVARIANT FUNCTOR $F: \mathbf{C} \rightarrow \mathbf{D}$ encodes two functions, one from $\text{obj}\mathbf{C}$ to $\text{obj}\mathbf{D}$ and one from $\text{mor}\mathbf{C}$ to $\text{mor}\mathbf{D}$, both denoted by $-^F$, such that:

- (i) the image of arrows of the form $a \xrightarrow{f} b$ in $\mathbf{C}$ is an arrow of the form $a^F \xleftarrow{f^F} b^F$ in $\mathbf{D}$;
- (ii) for every object c in $\mathbf{C}$, $(\text{id}_c)^F = \text{id}_{c^F}$;
- (iii) for any two composable arrows f and g , $(fg)^F = g^F f^F$.

A covariant functor $F: \mathbf{C}^{\text{op}} \rightarrow \mathbf{D}$ encodes two functions, one from $\text{obj}\mathbf{C}^{\text{op}}$ to $\text{obj}\mathbf{D}$ and one from $\text{mor}\mathbf{C}^{\text{op}}$ to $\text{mor}\mathbf{D}$, both denoted by $-^F$, such that:

- (i) the image of arrows of the form $a \xleftarrow{f^{\text{op}}} b$ in $\mathbf{C}^{\text{op}}$ is an arrow $a^F \xleftarrow{f^{\text{op}F}} b^F$ in $\mathbf{D}$;
- (ii) for every object c in $\mathbf{C}^{\text{op}}$, $(\text{id}_c)^F = \text{id}_{c^F}$;
- (iii) for any two composable arrows f^{op} and g^{op} , $(f^{\text{op}}g^{\text{op}})^F = f^{\text{op}F}g^{\text{op}F}$.

However, since every object and morphism of $\mathbf{C}^{\text{op}}$ is uniquely associated with one of $\mathbf{C}$, the preceding conditions can be reformulated without mentioning the opposite category, thereby obtaining the definition of contravariant functor. Thus, through a simple choice of the domain of the functor, one can implicitly indicate its *variance*, namely the feature of a functor of being covariant or contravariant; hence from this point on we shall indicate a contravariant functor F from $\mathbf{C}$ to $\mathbf{D}$ as $F: \mathbf{C}^{\text{op}} \rightarrow \mathbf{D}$. With this observation, many general results on covariant functors can be carried over to contravariant ones (for example, that every contravariant functor also sends isomorphisms to isomorphisms). We also record the following classical result, useful above all for handling the variance of a functor in the most convenient way for the context under consideration.

Proposition 1.4.8. *Every contravariant functor $F: \mathbf{C}^{\text{op}} \rightarrow \mathbf{D}$ corresponds bijectively to a covariant functor $G: \mathbf{C} \rightarrow \mathbf{D}^{\text{op}}$.*

Proof. Given the contravariant functor $F: \mathbf{C}^{\text{op}} \rightarrow \mathbf{D}$, consider a covariant functor $G: \mathbf{C} \rightarrow \mathbf{D}^{\text{op}}$ that acts on objects and identities as F does, and sends every arrow $a \xrightarrow{f} b$ of $\mathbf{C}$ to the arrow $a^F \xrightarrow{f^{\text{op}F}} b^F$ in $\mathbf{D}^{\text{op}}$ that uniquely corresponds to the arrow $a^F \xleftarrow{f^F} b^F$ in $\mathbf{D}$. The functor satisfies

$$(fg)^G = (fg)^{F^{\text{op}}} = (g^F f^F)^{\text{op}} = f^{F^{\text{op}}} g^{F^{\text{op}}} = f^G g^G$$

and is therefore well defined and encodes the data of the contravariant functor F . $\square$

Example 1.4.9. Given a linear map $U \xrightarrow{f} V$ between vector spaces over $\mathbb{K}$, and the corresponding dual spaces U^* and V^* , for every element $V \xrightarrow{\lambda} \mathbb{K}$ of V^* one can construct the map $U \xrightarrow{f\lambda} \mathbb{K}$, an element of U^* . Thus the dual-space functor can be constructed as a contravariant functor $\mathbf{Vect}_{\mathbb{K}}^{\text{op}} \xrightarrow{-^*} \mathbf{Vect}_{\mathbb{K}}$ that assigns to each space over $\mathbb{K}$ its dual, and to each morphism $U \xrightarrow{f} V$ between spaces a linear map given by precomposition, $V^* \xrightarrow{f-} U^*$.

$$\begin{array}{ccc} \mathbf{Vect}_{\mathbb{K}} & & \mathbf{Vect}_{\mathbb{K}} \\ U & & U^* \\ f \downarrow & \xrightarrow{-^*} & \uparrow f- \\ V & & V^* \end{array}$$

Example 1.4.10. A *presheaf* is a contravariant functor of the form $\mathbf{C}^{\text{op}} \xrightarrow{F} \mathbf{Set}$, a construction common in Topology and Algebraic Geometry. A *topological presheaf*, in particular, is a presheaf in which $\mathbf{C} = \text{open}(X)$ with X a topological space, i.e. $\text{open}(X)^{\text{op}} \xrightarrow{F} \mathbf{Set}$, and it sends every object U to a set U^F of functions s with domain U , called *sections*,

and such that if $s \in U^F$, then every restriction of it to an open subset W of U lies in W^F , while every inclusion $U \hookrightarrow V$ is sent to a restriction function $V^F \xrightarrow{\text{res}_{V,U}} U^F$ (which, by an abuse of notation, will always be denoted by res) that sends every section s of V to its restriction to U .

$$\begin{array}{ccc} \text{open}(X) & & \mathbf{Set} \\ U & \xrightarrow{F} & U^F \\ \downarrow & & \uparrow \text{res} \\ V & & V^F \end{array}$$

An example of a topological presheaf is the one that sends every open set U to the set of its sections with values in $\mathbb{R}$ (with the usual topology) that are C^∞ ; or, given a continuous map $A \xrightarrow{f} X$ between topological spaces, the one that assigns to U the set of continuous sections s such that the following diagram commutes:

$$\begin{array}{ccc} U & \xrightarrow{s} & A \\ & \searrow & \downarrow f \\ & & X \end{array}$$

Example 1.4.11. Given two categories $\mathbf{C}$ and $\mathbf{D}$, define their product $\mathbf{C} \times \mathbf{D}$ as the category whose objects are all pairs (c, d) with c and d objects respectively of $\mathbf{C}$ and $\mathbf{D}$, and whose morphisms $(c, d) \xrightarrow{(f,g)} (c', d')$ are all pairs of morphisms with $c \xrightarrow{f} c'$ in $\mathbf{C}$ and $d \xrightarrow{g} d'$ in $\mathbf{D}$, with componentwise composition. A *bifunctor* is a functor $\mathbf{A} \times \mathbf{B} \xrightarrow{F} \mathbf{C}$ whose domain is a product category. Writing $\mathbf{A} \times \mathbf{B}^{\text{op}} \xrightarrow{F} \mathbf{C}$ means that the functor is covariant in the first component and contravariant in the second, in accordance with the discussion following Definition 1.4.7. A functor of this kind, writing $c = (a, b)^F$ and $c' = (a', b')^F$, will be represented as

$$\begin{array}{ccc} \mathbf{A} \times \mathbf{B} & & \mathbf{C} \\ \begin{array}{cc} a & b \\ f \downarrow & \uparrow g \\ a' & b' \end{array} & \xrightarrow{F} & \begin{array}{c} c \\ \uparrow (f,g)^F \\ c' \end{array} \end{array}$$

It is easily verified that composition of functors, obtained from the usual composition of the respective two functions on objects and morphisms, is again a functor. A functor $F: \mathbf{C} \rightarrow \mathbf{D}$ is called an *isomorphism* between categories if there exists a functor $G: \mathbf{D} \rightarrow \mathbf{C}$ such that $FG = \text{id}_{\mathbf{D}}$ and $GF = \text{id}_{\mathbf{C}}$; in this case the two categories are said to be *isomorphic*, in symbols $\mathbf{C} \cong \mathbf{D}$.

Example 1.4.12. A Boolean algebra is a structure $(B, \wedge, \vee, \neg)$ in which $\vee$ and $\wedge$ are two binary operations on B and $\neg$ is a unary operation on B satisfying the following axioms for every $a, b, c \in B$:

- | | |
|--|---|
| (i) $(a \vee b) \vee c = a \vee (b \vee c)$
$(a \wedge b) \wedge c = a \wedge (b \wedge c)$ | (iv) $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$
$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$ |
| (ii) $a \vee b = b \vee a$
$a \wedge b = b \wedge a$ | (v) A neutral element 0 for $\vee$ and one 1 for $\wedge$ |
| (iii) $a \vee (a \wedge b) = a$
$a \wedge (a \vee b) = a$ | (vi) $a \vee \neg a = 1$
$a \wedge \neg a = 0$ |

The category **Bool** is the category whose objects are Boolean algebras and whose morphisms are Boolean algebra homomorphisms, that is, functions between the corresponding underlying sets that preserve all operations and the neutral elements. The full subcategory **Ring_B** of **Ring** whose objects are all unital rings R in which $a^2 = a \ \forall a \in R$ is isomorphic to **Bool**; indeed, the category is called the category of Boolean rings.

One can in fact verify that the following functors are inverse to each other.

$$\begin{array}{ccc}
 \mathbf{Bool} & & \mathbf{Ring}_B \\
 B & & (B, +_B, \cdot_B) \\
 f \downarrow & \xrightarrow{F} & \downarrow f^* \\
 C & & (C, +_C, \cdot_C)
 \end{array}$$

where the operations $\cdot_B$ and $+_B$ defined by $x \cdot_B y = x \wedge y$ and $x +_B y = (x \vee y) \wedge \neg(x \wedge y)$ for every $x, y \in B$ satisfy the axioms of rings with 1 and 0 as their respective neutral elements, while the arrow f^* acts pointwise as f , preserving the new operations and the neutral elements.

$$\begin{array}{ccc}
 \mathbf{Ring}_B & & \mathbf{Bool} \\
 R & & (R, \wedge_R, \vee_R, \neg_R) \\
 f \downarrow & \xrightarrow{G} & \downarrow f_* \\
 Q & & (Q, \wedge_Q, \vee_Q, \neg_Q)
 \end{array}$$

where the operations $\vee_R$, $\wedge_R$, and $\neg_R$ defined by $x \vee_R y = x \cdot y$, $x \wedge_R y = x + y + x \cdot y$, and $\neg_R x = 1 + x$ for every $x, y \in R$ satisfy the axioms of Boolean algebras with 1 and 0 as their respective neutral elements, while the arrow f_* acts pointwise as f , preserving the new operations and the neutral elements.

A functor $F: \mathbf{C} \rightarrow \mathbf{D}$ is called *full* when, for every two objects x and y of $\mathbf{C}$, every morphism $x^F \xrightarrow{f} y^F$ in $\mathbf{D}$ is the image of at least one morphism $x \xrightarrow{g} y$ in $\mathbf{C}$ (the functor is locally surjective on morphisms). The functor is instead called *faithful* when, for every two objects x and y of $\mathbf{C}$, the image of any morphism between them, $x^F \xrightarrow{f^F} y^F$ in $\mathbf{D}$,

comes from a unique morphism $x \xrightarrow{f} y$ (the functor is locally injective on morphisms). It is easy to prove that the composition of full functors is full and that of faithful functors is faithful.

Example 1.4.13. Categories such as **Set**, **Grp**, **Top**, **Man**, and **Vect** $_{\mathbb{K}}$ are called *concrete* because their objects and morphisms are sets and functions *equipped with some kind of structure*. The functor that sends every object to its underlying set and every morphism to the underlying function is called the *forgetful* functor $\mathbf{C} \xrightarrow{U} \mathbf{Set}$. In fact, it is precisely the presence of this functor that characterizes concrete categories: by definition, a category $\mathbf{C}$ is concrete if there exists a functor $\mathbf{C} \xrightarrow{U} \mathbf{Set}$ that is faithful. Indeed, every set can be equipped with many structures or none, and hence can correspond to many or no objects of $\mathbf{C}$, so there is no condition of surjectivity or injectivity on objects, and even less on morphisms. On the other hand, however, to every morphism of the category there must correspond a single function, but it is not the case that any two functions between two sets induce some structure-preserving morphism between the corresponding objects of the category. Thus forgetful functors, such as $\mathbf{Grp} \xrightarrow{U} \mathbf{Set}$, are not necessarily full. Functors that *forget part of the structure* are also often called forgetful functors; for example, the functor $\mathbf{Ring} \xrightarrow{U} \mathbf{Ab}$ that assigns to each unital ring the corresponding additive group and to each ring morphism the evident homomorphism between abelian groups is referred to as a forgetful functor.

Given an object c of a locally small category $\mathbf{C}$, the *covariant functor represented by c* is the functor described as follows:

$$\begin{array}{ccc} \mathbf{C} & & \mathbf{Set} \\ x & & \mathbf{C}(c, x) \\ f \downarrow & \xrightarrow{\mathbf{C}(c, -)} & \downarrow -f \\ y & & \mathbf{C}(c, y) \end{array}$$

whereas the *contravariant functor represented by c* is the functor

$$\begin{array}{ccc} \mathbf{C} & & \mathbf{Set} \\ x & & \mathbf{C}(x, c) \\ f \downarrow & \xrightarrow{\mathbf{C}(-, c)} & \uparrow f- \\ y & & \mathbf{C}(y, c) \end{array}$$

The morphisms $-f$ and $f-$ are called postcomposition and precomposition by f , and are functions described in the following way:

$$-f: g \in \mathbf{C}(c, x) \mapsto fg \in \mathbf{C}(c, y)$$

$$f-: g \in \mathbf{C}(x, c) \mapsto gf \in \mathbf{C}(y, c)$$

Obviously, the requirement that the category $\mathbf{C}$ be locally small has to do with the well-definedness of the functor, but as discussed in the introductory paragraph, this requirement, like other similar ones that will be made later, does not prevent a generalization whenever this is allowed by a different foundation for category theory.

1.5 Naturality

Definition 1.5.1. Given two functors $F, G: \mathbf{C} \rightarrow \mathbf{D}$ between the categories $\mathbf{C}$ and $\mathbf{D}$, a NATURAL TRANSFORMATION is a function $\eta: c \in \text{obj}\mathbf{C} \mapsto \eta_c \in \text{mor}\mathbf{D}$ such that the following diagram commutes for every morphism $a \xrightarrow{f} b$ in $\mathbf{C}$:

$$\begin{array}{ccc} a^F & \xrightarrow{\eta_a} & a^G \\ f^F \downarrow & & \downarrow f^G \\ b^F & \xrightarrow{\eta_b} & b^G \end{array}$$

The transformation will be denoted by $\eta: F \Rightarrow G$, or also by

$$\begin{array}{ccc} & F & \\ \text{C} & \begin{array}{c} \curvearrowright \\ \eta \\ \curvearrowleft \end{array} & \text{D} \\ & G & \end{array}$$

When each η_c is an isomorphism, η is called a *natural isomorphism* and the functors are said to be *isomorphic*, in symbols $F \cong G$. In particular, for every object x of $\mathbf{C}$, one says that x^F and x^G are *naturally isomorphic*.

Example 1.5.2. Consider the category $\mathbf{Vect}_{\mathbb{K}}^{\text{fin}}$ and its two endofunctors

$$\text{id}_{\mathbf{Vect}_{\mathbb{K}}^{\text{fin}}}, -^{**}: \mathbf{Vect}_{\mathbb{K}}^{\text{fin}} \rightarrow \mathbf{Vect}_{\mathbb{K}}^{\text{fin}}$$

namely the identity functor and the double-dual functor. Indeed, by Proposition 1.4.8 the dual functor $-^*: \mathbf{Vect}_{\mathbb{K}}^{\text{fin op}} \rightarrow \mathbf{Vect}_{\mathbb{K}}^{\text{fin}}$ can be seen as a functor $\mathbf{Vect}_{\mathbb{K}}^{\text{fin}} \rightarrow \mathbf{Vect}_{\mathbb{K}}^{\text{fin op}}$, and we shall continue to denote it by $-^*$; hence the double-dual functor is defined as the composition of two dual functors, the first viewed as a covariant functor and the second as a contravariant functor:

$$\mathbf{Vect}_{\mathbb{K}}^{\text{fin}} \xrightarrow{-^*} \mathbf{Vect}_{\mathbb{K}}^{\text{fin op}} \xrightarrow{-^*} \mathbf{Vect}_{\mathbb{K}}^{\text{fin}}$$

Consider the following natural transformation:

$$\begin{array}{ccc}
 & \text{id}_{\mathbf{Vect}_{\mathbb{K}}^{\text{fin}}} & \\
 \text{Vect}_{\mathbb{K}}^{\text{fin}} & \begin{array}{c} \downarrow \text{ev} \\ \downarrow \end{array} & \text{Vect}_{\mathbb{K}}^{\text{fin}} \\
 & \text{--}^{**}\text{--} &
 \end{array}$$

The transformation ev has as components the isomorphisms ev_V for every object V in $\mathbf{Vect}_{\mathbb{K}}$, which, for brevity and as is usually done, will be denoted simply by ev regardless of the space:

$$V \xrightarrow{\text{ev}} V^{**}$$

it assigns to each vector v of the domain the map $v^{\text{ev}} : \lambda \in V^* \mapsto v\lambda \in \mathbb{K}$, which assigns to each λ its evaluation at v . Thus every vector space is naturally isomorphic to its double dual. It is in fact well known that every finite-dimensional space is isomorphic to its dual and hence to the dual of its dual: it is therefore natural to ask whether naturality also holds between a space and its dual. The answer is negative, for the simple reason that the identity functor and the dual functor do not have the same variance. If one wishes to be more permissive, there is a concept of *extranaturality* allowing one to discuss the naturality of a transformation independently of variance. However, even extranaturality fails⁵: if the transformation were extranatural, then the isomorphism between U and U^* would have to decompose as follows for every morphism ϕ :

$$\begin{array}{ccc}
 U & \xrightarrow{\text{ev}} & U^* \\
 \phi \downarrow & & \uparrow \phi- \\
 V & \xrightarrow{\text{ev}} & V^*
 \end{array}$$

but then it is easy to check that every ϕ would have to be a monomorphism, which is obviously false if $\dim U > 0$.

Example 1.5.3. Given two objects x and y of the locally small category $\mathbf{C}$, any morphism $x \xrightarrow{f} y$ induces a natural transformation between the functors, of the appropriate variance, represented by the two objects; for example:

$$\begin{array}{ccc}
 & \mathbf{C}(-, x) & \\
 \mathbf{C} & \begin{array}{c} \downarrow \eta \\ \downarrow \end{array} & \mathbf{Set} \\
 & \mathbf{C}(-, y) &
 \end{array}$$

where η assigns to each object a the morphism $\eta_a = -f$. Thus, for every morphism $a \xrightarrow{g} b$ of $\mathbf{C}$, by the naturality of η one has:

⁵A proof by Eilenberg and Mac Lane can be found in [EM45] (p. 234).

$$\begin{array}{ccc}
\mathbf{C}(a, x) & \xrightarrow{-f} & \mathbf{C}(a, y) \\
\downarrow g- & & \downarrow g- \\
\mathbf{C}(b, x) & \xrightarrow{-f} & \mathbf{C}(b, y)
\end{array}$$

Example 1.5.4. Consider the functors $\mathrm{GL}_n, -^\times : \mathbf{CRing} \rightarrow \mathbf{Grp}$. The first sends every ring to the group of invertible matrices over it of size n , and every ring morphism to the group morphism that transforms the entries of the matrix by means of the original morphism; the second sends every ring to its group of units and every morphism to the group morphism induced by restriction. One obtains a natural transformation

$$\begin{array}{ccc}
& \mathrm{GL}_n & \\
\mathrm{CRing} & \begin{array}{c} \downarrow \det \\ \downarrow \end{array} & \mathbf{Grp} \\
& -^\times &
\end{array}$$

which assigns to every ring R the morphism $\det_R : M \in \mathrm{GL}_n(R) \mapsto \det(M) \in R^\times$, which we shall therefore denote directly by $\det$:

$$\begin{array}{ccc}
\mathrm{GL}_n(R) & \xrightarrow{\det} & R^\times \\
\downarrow f^* & & \downarrow f_* \\
\mathrm{GL}_n(K) & \xrightarrow{\det} & K^\times
\end{array}$$

this is true because the determinant is an invariant of endomorphisms, that is, the value of the determinant of the matrix depends only on the endomorphism represented by the matrix and not on the particular matrix: computing the determinant and then applying f is the same as applying f to the coefficients and then computing the determinant.

Example 1.5.5. In general, given two categories $\mathbf{C}$ and $\mathbf{D}$, the category of functors $\mathbf{C} \rightarrow \mathbf{D}$ will be denoted by $\mathbf{D}^{\mathbf{C}}$. Its objects are, precisely, all functors of the form $\mathbf{C} \rightarrow \mathbf{D}$, and its morphisms are natural transformations between functors, which are composed component by component.

One must also mention a particular type of *mixed* composition between functors and natural transformations, called *whiskering*. Consider categories, functors, and natural

transformations as in the figure:

$$\mathbf{C} \xrightarrow{F} \mathbf{D} \begin{array}{c} \xrightarrow{G} \\ \eta \\ \xrightarrow{H} \end{array} \mathbf{A} \xrightarrow{L} \mathbf{B}$$

we define the natural transformation

$$\mathbf{C} \begin{array}{c} \xrightarrow{FG} \\ F\eta \\ \xrightarrow{FH} \end{array} \mathbf{A}$$

as the one whose component is $(F\eta)_c = \eta_{c^F}$ for every object c of $\mathbf{C}$; indeed, from the naturality of η follows the commutativity of the following diagram, and hence the naturality of $F\eta$:

$$\begin{array}{ccc} x^{FG} & \xrightarrow{\eta_{x^F}} & x^{FH} \\ f^{FG} \downarrow & & \downarrow f^{FH} \\ y^{FG} & \xrightarrow{\eta_{y^F}} & y^{FH} \end{array}$$

Instead, we define the natural transformation

$$\mathbf{D} \begin{array}{c} \xrightarrow{GL} \\ \eta^L \\ \xrightarrow{FL} \end{array} \mathbf{B}$$

as the one whose component is $(\eta^L)_d = \eta_d^L$ for every object d of $\mathbf{D}$; indeed, from the naturality of η follows the commutativity of the diagram on the left, and applying the functor L gives the commutativity of the one on the right:

$$\begin{array}{ccc} x^G & \xrightarrow{\eta_x} & x^H \\ f^G \downarrow & & \downarrow f^H \\ y^G & \xrightarrow{\eta_y} & y^H \end{array} \quad \begin{array}{ccc} x^{GL} & \xrightarrow{\eta_x^L} & x^{HL} \\ f^{GL} \downarrow & & \downarrow f^{HL} \\ y^{GL} & \xrightarrow{\eta_y^L} & y^{HL} \end{array}$$

It is clear that, if the functor F or L had been parallel to G and H , one could have composed on both sides of η . Finally, we add the following proposition, whose verification is straightforward.

Proposition 1.5.6. *The following properties hold for whiskering:*

(i) *In the notation above,*

$$\eta = \text{id}_{\mathbf{D}} \eta = \eta \text{id}_{\mathbf{A}} \quad \text{id}_F G = \text{id}_{FG} \quad F \text{id}_G = \text{id}_{FG}$$

(ii) *If F, F', G, G' , and η are as in the figure*

$$\mathbf{C}' \xrightarrow{F'} \mathbf{C} \xrightarrow{F} \mathbf{A} \begin{array}{c} \xrightarrow{H} \\ \Downarrow \eta \\ \xrightarrow{L} \end{array} \mathbf{B} \xrightarrow{G} \mathbf{D} \xrightarrow{G'} \mathbf{D}'$$

then the following hold:

$$(F\eta)G = F(\eta G) \quad F'(F\eta) = (F'F)\eta \quad \eta(GG') = (\eta G)G'$$

(iii) *If F, G, η , and ε are as in the figure*

$$\mathbf{C} \xrightarrow{F} \mathbf{A} \begin{array}{c} \xrightarrow{H} \\ \Downarrow \eta \\ \Downarrow \varepsilon \\ \xrightarrow{L} \end{array} \mathbf{B} \xrightarrow{G} \mathbf{D}$$

then the following hold:

$$F(\eta\varepsilon) = (F\eta)(F\varepsilon) \quad (\eta\varepsilon)G = (\eta G)(\varepsilon G)$$

Definition 1.5.7. Two categories $\mathbf{C}$ and $\mathbf{D}$ are EQUIVALENT if there exist two functors F and G such that $FG \cong \text{id}_{\mathbf{D}}$ and $GF \cong \text{id}_{\mathbf{C}}$.

Example 1.5.8. The category $\mathbf{Mat}_{\mathbb{K}}$, as presented in Example 1.1.6, is equivalent to the full subcategory $\mathbf{Vect}_{\mathbb{K}}^{\text{fin}}$ of $\mathbf{Vect}_{\mathbb{K}}$. Fix a basis for every space of $\mathbf{Vect}_{\mathbb{K}}^{\text{fin}}$, and in particular fix the canonical basis for the numerical vector spaces $\mathbb{K}^n$. Once this choice has been made, one can define the functor

$$\begin{array}{ccc} \mathbf{Vect}_{\mathbb{K}}^{\text{fin}} & & \mathbf{Mat}_{\mathbb{K}} \\ \begin{array}{c} V \\ f \downarrow \\ U \end{array} & \xrightarrow{\quad \text{dim} \quad} & \begin{array}{c} \text{dim}_{\mathbb{K}} V \\ \downarrow M_f \\ \text{dim}_{\mathbb{K}} U \end{array} \end{array}$$

where M_f is the matrix representing the homomorphism f with respect to the chosen bases. Consider also the functor

$$\begin{array}{ccc}
\mathbf{Mat}_{\mathbb{K}} & & \mathbf{Vect}_{\mathbb{K}}^{\text{fin}} \\
\begin{array}{c} n \\ A \downarrow \\ m \end{array} & \xrightarrow{\mathbb{K}^-} & \begin{array}{c} \mathbb{K}^n \\ \downarrow \text{map}_A \\ \mathbb{K}^m \end{array}
\end{array}$$

where map_A is the homomorphism represented by A from the numerical space $\mathbb{K}^n$ to $\mathbb{K}^m$ with respect to the canonical bases. For every space V of $\mathbf{Vect}_{\mathbb{K}}^{\text{fin}}$ with $\dim V = n$, consider the isomorphism $V \xrightarrow{\phi_V} \mathbb{K}^n$ due to the chosen basis; these isomorphisms, as V ranges over $\mathbf{Vect}_{\mathbb{K}}^{\text{fin}}$, assemble into a natural transformation $\phi: \dim \mathbb{K}^- \Rightarrow \text{id}_{\mathbf{Vect}_{\mathbb{K}}^{\text{fin}}}$, since the following diagram is commutative for every arrow $V \xrightarrow{f} U$, where $\dim U = m$ and A is the matrix representing f :

$$\begin{array}{ccc}
V & \xrightarrow{\phi_V} & \mathbb{K}^n \\
\downarrow f & & \downarrow \text{map}_A \\
U & \xrightarrow{\phi_U} & \mathbb{K}^m
\end{array}$$

On the other hand, by the way the bases were fixed, $\mathbb{K}^- \dim = \text{id}_{\mathbf{Mat}_{\mathbb{K}}}$, and hence the categories $\mathbf{Mat}_{\mathbb{K}}$ and $\mathbf{Vect}_{\mathbb{K}}^{\text{fin}}$ are equivalent.

Chapter 2

Universality

2.1 Universal arrows

Definition 2.1.1. An object c of a category $\mathbf{C}$ is called **INITIAL** if and only if, for every object x of $\mathbf{C}$, there exists a unique arrow from c to x . Dually, c is called **TERMINAL** if and only if, for every object x of $\mathbf{C}$, there exists a unique arrow from x to c .

Initial objects are unique up to isomorphism: if a and b are initial, then there is a unique arrow $a \xrightarrow{f} b$ and a unique arrow $b \xrightarrow{g} a$, but since the arrow from a to a is unique, $fg = 1_a$; analogously one has $gf = 1_b$. Dually, terminal objects are also unique up to isomorphism.

If an object is both initial and terminal, it is called a *zero object* of the category. If z is a zero object of the category $\mathbf{C}$, then for any two objects a and b in $\mathbf{C}$, considering the arrows $a \xrightarrow{f} z$ and $z \xrightarrow{g} b$, their composite fg is called a *zero arrow* from a to b . It is easy to prove that this arrow is unique.

Example 2.1.2. The empty set in **Set** is initial because for every set there exists an empty function having it as codomain, and this function is trivially unique. Again in **Set**, every singleton is terminal: indeed, singletons are all isomorphic (in bijection), because for every set there is a unique function mapping each of its elements to the unique element of the singleton.

Example 2.1.3. In the category **Rng**, as presented in Example 1.1.9, the zero ring is terminal and initial, that is, it is the zero object of the category. However, in **Ring** it is only terminal; indeed, there are no morphisms at all having the zero ring as domain and a nonzero ring as codomain, since ring homomorphisms from this ring do not preserve the unit. Instead, the ring $\mathbb{Z}$ is initial, since the unique unital ring homomorphism from $\mathbb{Z}$ to a ring R is the homomorphism $n \in \mathbb{Z} \mapsto ne \in R$, with e the unit of R .

Example 2.1.4. The trivial group is the zero object of **Grp** and **Ab**.

Definition 2.1.5. A functor $F: \mathbf{C} \rightarrow \mathbf{Set}$ from a locally small category $\mathbf{C}$ is called **REPRESENTABLE** if it is naturally isomorphic to the functor represented, with the appropriate

variance, by some object c of $\mathbf{C}$, which we shall call a *representing object* of F .

With the notation used above, we shall say that F is *represented* by c , whereas by a *representation* of F we shall mean a natural isomorphism from F to the functor represented by c .

Example 2.1.6. The identity functor $\text{id}_{\mathbf{Set}}: \mathbf{Set} \rightarrow \mathbf{Set}$ is represented by a singleton, let it be the usual $*$ $= \{p\}$: we shall see that representing objects are unique up to isomorphism in Proposition 2.1.23. Indeed, consider the representation $\phi: \mathbf{Set}(*, -) \rightarrow \text{id}_{\mathbf{Set}}$, whose components are $\phi_A: * \xrightarrow{a} A \in \mathbf{Set}(*, -) \mapsto a \in A$, where $* \xrightarrow{a} A$ is the function sending p to a . We shall often identify the elements $a \in A$ with the arrows $* \xrightarrow{a} A$.

Example 2.1.7. The ring of Laurent polynomials over $\mathbb{Z}$ is the ring $\mathbb{Z}[x, x^{-1}]$. One can show that the only morphisms in $\mathbf{Ring}$ from this ring to a generic ring R are the evaluation maps v_u that evaluate polynomials at an invertible element u of R . Therefore a representation of $-^\times: \mathbf{Ring} \rightarrow \mathbf{Grp}$ (composed with the forgetful functor U , as in Example 1.4.13) is given by $\phi: \mathbf{Ring}(\mathbb{Z}[x, x^{-1}], -) \Rightarrow -^\times U$, with natural components

$$\phi_R: \mathbb{Z}[x, x^{-1}] \xrightarrow{v} R \in \mathbf{Ring}(\mathbb{Z}[x, x^{-1}], R) \mapsto x^v \in (R^\times)^U$$

Example 2.1.8. The contravariant functor $\mathcal{P}: \mathbf{Set}^{\text{op}} \rightarrow \mathbf{Set}$ is represented by the set 2; indeed, the power set of any set is naturally isomorphic to its set of functions into 2, namely the characteristic functions of its subsets.

Definition 2.1.9. Given a functor $F: \mathbf{D} \rightarrow \mathbf{C}$ and an object c of $\mathbf{C}$, a **UNIVERSAL ARROW** from c to F is a pair (u, x) in which x is an object of $\mathbf{D}$ and $c \xrightarrow{u} x^F$ is an arrow of $\mathbf{C}$ such that, for every arrow of $\mathbf{C}$ of the form $c \xrightarrow{v} d^F$ with d in $\mathbf{D}$, there exists a unique $x \xrightarrow{f} d$ in $\mathbf{D}$ such that $uf^F = v$:

$$\begin{array}{ccc} c & \xrightarrow{u} & x^F \\ & \searrow v & \downarrow f^F \\ & & d^F \end{array} \qquad \begin{array}{c} x \\ \downarrow !f \\ d \end{array}$$

We shall often refer imprecisely to the arrow u as the universal arrow, leaving implicit the object x in question. Let $\mathbf{1}$ denote the category consisting of a single object, call it p , and a single morphism, namely the identity. Given an object c of the category $\mathbf{C}$, we shall identify c with the functor $c: \mathbf{1} \rightarrow \mathbf{C}$ that sends the object of $\mathbf{1}$ to c (and hence the morphism of $\mathbf{1}$ to id_c). With the notation above and that of Example 1.4.6, the category $c \downarrow F$ has as objects triples (a, f, p) , which we shall identify with pairs (f, a) , since the third component is forced, as a ranges over the objects of $\mathbf{D}$ and f over the arrows in $\mathbf{C}(c, a^F)$. After this substitution, the morphisms must be of the form $(f, a) \xrightarrow{(h, \text{id}_p)} (g, b)$

and hence may in turn be replaced by $(f, a) \xrightarrow{h} (g, b)$, where $a \xrightarrow{h} b$ is a morphism of $\mathbf{D}$, but it will be identified with the corresponding morphism of the comma category, which makes the following diagram commute:

$$\begin{array}{ccc} c & \xrightarrow{f} & a^F \\ & \searrow g & \downarrow h^F \\ & & b^F \end{array}$$

Equivalently, one can say that a universal arrow (u, x) from c to F is an initial object in $c \downarrow F$. It follows directly that a universal arrow is unique up to isomorphism. For later use, this category will be called the *slice* category of c over F and denoted by $\mathbf{c}/\mathbf{F}$. Moreover, if F is the identity functor of $\mathbf{C}$, the slice category will be denoted by $\mathbf{c}/\mathbf{C}$.

Example 2.1.10. Consider the functor $L : \mathbf{Set} \rightarrow \mathbf{Vect}_{\mathbb{K}}$ that assigns to every set S a chosen vector space S^L over $\mathbb{K}$ with basis S , and to every morphism $S \xrightarrow{f} T$ the unique homomorphism that acts as f on the basis S of S^L . Also consider the forgetful functor $U : \mathbf{Vect}_{\mathbb{K}} \rightarrow \mathbf{Set}$. The inclusion $S \xrightarrow{i} S^{LU}$ is universal from S to U ; indeed, for every arrow $S \xrightarrow{f} V^U$ there exists a unique $S^L \xrightarrow{\bar{f}} V$ making the following diagram commute, and it is a classical result of linear algebra that $\bar{f}$ is precisely the extension of f by $\mathbb{K}$ -linearity:

$$\begin{array}{ccc} S & \xrightarrow{i} & S^{LU} \\ & \searrow f & \downarrow \bar{f}^U \\ & & V^U \end{array} \qquad \begin{array}{ccc} S^L & & \\ & \downarrow \bar{f} & \\ & & V \end{array}$$

Example 2.1.11. Let R be a commutative ring having a field $\mathbb{K}$ as a subring; it may therefore be regarded as a $\mathbb{K}$ -algebra. A derivation on R is a function $R \xrightarrow{D} M$, with M an R -module, which is $\mathbb{K}$ -linear and such that $D(xy) = D(x)y + xD(y)$ for every $x, y \in R$. Consider the functor $D_R : \mathbf{Mod}_R \rightarrow \mathbf{Set}$ that assigns to each R -module M its set of derivations of the form $R \xrightarrow{D} M$, and to each morphism $M \xrightarrow{f} N$ between R -modules the postcomposition function $-f$, viewing f as a morphism between $\mathbb{K}$ -modules. If R has underlying set S , we shall make a formal distinction by denoting each element x of R by $\bar{x}$ when viewed as an element of S . Let F then be the free R -module generated by S , and consider its submodule N generated by the set

$$\{ \overline{x+y} - (\bar{x} + \bar{y}) , \ \overline{\lambda x} - \lambda \bar{x} , \ \overline{xy} - (\bar{x}y + x\bar{y}) \mid x, y \in R , \ \lambda \in \mathbb{K} \}$$

The quotient $\Omega_{\mathbb{K}}(R) := F/N$ will be called the $\mathbb{K}$ -module of differentials of R , and to it one associates a derivation $R \xrightarrow{d} \Omega_{\mathbb{K}}(R)$ such that $d(x) = \pi(\bar{x})$ for every $x \in R$, where π is the canonical projection from F onto $\Omega_{\mathbb{K}}(R)$. One can prove that if $R \xrightarrow{D} M$ is any derivation on R , then D factors uniquely as $D = d\phi$, with $\Omega_{\mathbb{K}}(R) \xrightarrow{\phi} M$ a morphism of R -modules.

$$\begin{array}{ccc} R & \xrightarrow{d} & \Omega_{\mathbb{K}}(R) \\ & \searrow D & \downarrow \phi \\ & & M \end{array}$$

This can be formalized by saying that $(d, \Omega_{\mathbb{K}}(R))$ is universal from $*$ to D_R .

$$\begin{array}{ccccc} * & \xrightarrow{d} & \Omega_{\mathbb{K}}(R)^{D_R} & & \Omega_{\mathbb{K}}(R) \\ & \searrow D & \downarrow -\phi & & \downarrow \phi \\ & & M^{D_R} & & M \end{array}$$

It is often said, also in other similar situations, that d is universal among the derivations on R .

Proposition 2.1.12. *Let $\mathbf{C}$ and $\mathbf{D}$ be locally small categories. Given a functor $F: \mathbf{D} \rightarrow \mathbf{C}$ and an object c of $\mathbf{C}$, consider an object x of $\mathbf{D}$ and an arrow $c \xrightarrow{u} x^F$ of $\mathbf{C}$. Let $\hat{u} = \eta$ be a family of functions $\eta = (\eta_d)_{d \in \text{obj } \mathbf{D}}$ where, for every object d in $\mathbf{D}$,*

$$\eta_d: x \xrightarrow{f} d \in \mathbf{D}(x, d) \mapsto uf^F \in \mathbf{C}(c, d^F)$$

Then η is a natural transformation

$$\begin{array}{ccc} & \mathbf{D}(x, -) & \\ \curvearrowright & \parallel \eta & \curvearrowright \\ \mathbf{D} & & \mathbf{Set} \\ \curvearrowleft & & \curvearrowleft \\ & \mathbf{C}(c, -^F) & \end{array}$$

and moreover:

(i) *if N is the class of natural transformations from $\mathbf{D}(x, -)$ to $\mathbf{C}(c, -^F)$, then*

$$T: u \in \mathbf{C}(c, x^F) \mapsto \hat{u} \in N$$

is a bijection whose inverse is the function

$$S: \eta \in N \mapsto 1_x^{\eta_x} \in \mathbf{C}(c, x^F)$$

(ii) for every $u \in \mathbf{C}(c, x^F)$, $\hat{u}$ is an isomorphism if and only if (u, x) is a universal arrow from c to F ; that is, T induces a bijection from the class of natural isomorphisms in N to the class of universal arrows from c to F .

Proof. Given an arbitrary arrow $a \xrightarrow{\phi} b$ of $\mathbf{D}$, the diagram

$$\begin{array}{ccc} \mathbf{D}(x, a) & \xrightarrow{\eta_a} & \mathbf{C}(c, a^F) \\ \downarrow -\phi & & \downarrow -\phi^F \\ \mathbf{D}(x, b) & \xrightarrow{\eta_b} & \mathbf{C}(c, b^F) \end{array}$$

expresses the naturality of η ; indeed, for any arrow $x \xrightarrow{f} a$ in $\mathbf{D}(x, a)$, one has:

$$f^{(-\phi)\eta_b} = (f\phi)^{\eta_b} = u(f\phi)^F = uf^F\phi^F = (uf^F)^{-\phi^F} = f\eta_a(-\phi^F)$$

as illustrated in the following diagram:

$$\begin{array}{ccccc} c & \xrightarrow{u} & x^F & & x \\ & \searrow & \downarrow f^F & & \downarrow f \\ & & a^F & & a \\ & \searrow & \downarrow \phi^F & & \downarrow \phi \\ & & b^F & & b \end{array}$$

(i) Consider a natural transformation $\eta : \mathbf{D}(x, -) \Rightarrow \mathbf{C}(c, -^F)$. If 1_x is the identity of x , then it is an element of $\mathbf{D}(x, x)$, and hence is sent by η_x to an arrow u of $\mathbf{C}(c, x^F)$. For any arrow $x \xrightarrow{f} d$ of $\mathbf{D}$, one has the commutative diagram

$$\begin{array}{ccc} \mathbf{D}(x, x) & \xrightarrow{\eta_x} & \mathbf{C}(c, x^F) \\ \downarrow -f & & \downarrow -f^F \\ \mathbf{D}(x, d) & \xrightarrow{\eta_d} & \mathbf{C}(c, d^F) \end{array}$$

Focusing on how 1_x is transformed, one obtains:

$$f^{\eta_d} = (1_x f)^{\eta_d} = 1_x^{(-f)\eta_d} = 1_x^{\eta_x(-f^F)} = (1_x^{\eta_x})^{-f^F} = u^{-f^F} = u f^F$$

that is, $\eta_d: f \mapsto u f^F$, hence $\eta = \hat{u}$, and we can therefore write $\eta^{ST} = u^T = \hat{u} = \eta$. On the other hand, $u^{TS} = \hat{u}^S = 1_x^{\hat{u}_x} = u 1_x^F = u 1_{x^F} = u$, so indeed $S = T^{-1}$.

(ii) Saying that η is an isomorphism means that each component η_d is a bijection, but this follows directly from the definition of universal arrow; in fact it is merely another way of giving the definition. $\square$

Thus x is a representing object for $\mathbf{C}(c, -^F)$, and any representation of $\mathbf{C}(c, -^F)$ is determined by a universal arrow from c to F .

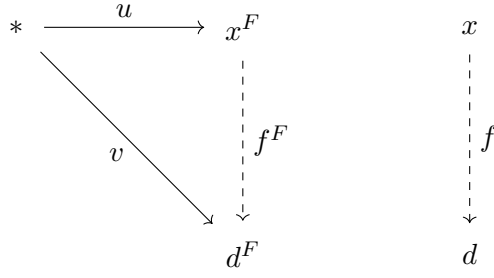
Definition 2.1.13. Given a category $\mathbf{D}$ and a functor $F: \mathbf{D} \rightarrow \mathbf{Set}$, a **UNIVERSAL ELEMENT** of F is a pair (u, x) , where x is an object of $\mathbf{D}$ and $u \in x^F$, such that for every pair (v, d) with $v \in d^F$ there exists a unique arrow $x \xrightarrow{f} d$ of $\mathbf{D}$ such that $u f^F = v$.

Example 2.1.14. Given a group G and a normal subgroup N of it, one can give an alternative description of the quotient G/N . Let $\text{Nih}_{G/N}: \mathbf{Grp} \rightarrow \mathbf{Set}$ be the functor that assigns to every group H the set of homomorphisms $G \xrightarrow{f} H$ that annihilate N , namely $N \xrightarrow{\vec{f}} 1$, while to each morphism $H \xrightarrow{\phi} K$ it assigns the map defined by postcomposition with ϕ from $H^{\text{Nih}_{G/N}}$ to $K^{\text{Nih}_{G/N}}$. The pair $(\pi, G/N)$, where $G \xrightarrow{\pi} G/N$ is the canonical projection, is a universal element of $\text{Nih}_{G/N}$; indeed, for every morphism $G \xrightarrow{f} H$ in $H^{\text{Nih}_{G/N}}$ there exists a unique $G/N \xrightarrow{\bar{f}} H$ induced by f on the quotient such that

$$\pi \bar{f} = f$$

Example 2.1.15. Given vector spaces V and U over the field $\mathbb{K}$, consider the functor $\text{Bil}_{U,V}: \mathbf{Vect}_{\mathbb{K}} \rightarrow \mathbf{Set}$ that assigns to each space W the set of bilinear functions $U \times V \rightarrow W$ and acts on morphisms by postcomposition. The bilinear function $\otimes: (u, v) \in U \times V \mapsto u \otimes v \in U \otimes V$ is a universal element for $\text{Bil}_{U,V}$ because every other bilinear function factors through it.

The concept of universal element is in fact interchangeable with that of universal arrow. With the notation of the preceding definition, given the universal element (u, x) , one can identify $u \in x^F$ with the arrow $* \xrightarrow{u} x^F$ in $\mathbf{Set}$, and similarly for v in (v, d) ; what is required of these elements and of the morphism f translates into the commutativity of the following diagram:



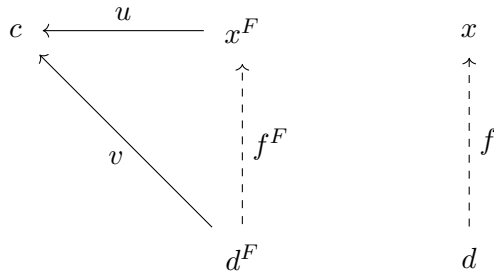
and thus (u, x) can be interpreted as a universal arrow from $*$ to F .

Example 2.1.16. Returning to Example 2.1.11, one deduces that that arrow is in fact interpretable as a universal element. As in this example and in the other examples concerning universal elements of a functor $F: \mathbf{C} \rightarrow \mathbf{Set}$, the elements of the sets c^F are often arrows, and one says that the universal element is universal among the arrows of c^F for every c .

Conversely, given a universal arrow (u, x) from c to $F: \mathbf{D} \rightarrow \mathbf{C}$, the arrow $c \xrightarrow{u} x^F$ is an element of $\mathbf{C}(c, x^F)$, and every arrow of the form $c \xrightarrow{v} d^F$, with d an object of $\mathbf{D}$, is an element of $\mathbf{C}(c, d^F)$; thus there exists a unique arrow $x \xrightarrow{f} d$ such that $uf^F = v$. This induces a postcomposition function $\mathbf{C}(c, x^F) \xrightarrow{-f^F} \mathbf{C}(c, d^F)$ such that $u^{-f^F} = v$, and therefore (u, x) can in turn be viewed as a universal element of $\mathbf{C}(c, -^F)$.

Hence a universal arrow (u, x) from c to F corresponds to a universal element (u, x) of $\mathbf{C}(c, -^F)$ and to a representation of the functor $\mathbf{C}(c, -^F)$. Conversely, a universal element (u, x) , or a representation of $\mathbf{C}(c, -^F)$, corresponds to a universal arrow, in the ways already observed.

Definition 2.1.17. Given a functor $F: \mathbf{D} \rightarrow \mathbf{C}$ and an object c of $\mathbf{C}$, a **UNIVERSAL ARROW** from F to c is a pair (u, x) in which x is an object of $\mathbf{D}$ and $x^F \xrightarrow{u} c$ is an arrow of $\mathbf{C}$ such that, for every arrow of $\mathbf{C}$ of the form $d^F \xrightarrow{v} c$ with d in $\mathbf{D}$, there exists a unique $d \xrightarrow{f} x$ in $\mathbf{D}$ making the following diagram commute:



Clearly this is the dual concept of a universal arrow from an object to a functor. By dualizing, indeed, one obtains that a universal arrow (u, x) from F to c is a terminal object in $c \downarrow F$. As before, this will be called the *slice* category of F over c , denoted by $\mathbf{F}/c$, and if F is the identity functor of $\mathbf{C}$, abbreviated as $\mathbf{C}/c$.

Proposition 2.1.18. *Let $F: \mathbf{D} \rightarrow \mathbf{Set}$ be a functor with $\mathbf{D}$ locally small, and let (u, x) be a universal arrow from $*$ to F (that is, a universal element of F). For every object d of $\mathbf{D}$, consider the function:*

$$\psi_d : x \xrightarrow{f} d \in \mathbf{D}(x, d) \mapsto p^{u f^F} \in d^F$$

These assemble into a natural isomorphism, and hence provide a representation of F . Moreover, every representation of such an F has this form.

Proof. For every set X , define the bijection:

$$\eta_X : * \xrightarrow{x} X \in \mathbf{Set}(*, X) \mapsto p^x \in X$$

Functions of this kind assemble into a natural isomorphism

$$\eta : \mathbf{Set}(*, -) \Rightarrow \text{id}_{\mathbf{Set}}$$

By postcomposing these functors with the functor F , we obtain a natural isomorphism

$$\bar{\eta} : \mathbf{Set}(*, -^F) \Rightarrow F$$

The proposition therefore follows from Proposition 2.1.12, since the representations of $\mathbf{Set}(*, -^F)$ are exactly those described in that way. $\square$

Before stating one of the most important results of Category Theory, we list some notation. Given a functor $H : \mathbf{D} \rightarrow \mathbf{Set}$ with $\mathbf{D}$ locally small and an object x in $\mathbf{D}$, the class of natural transformations from $\mathbf{D}(x, -)$ to H will be denoted by $\text{Nat}(\mathbf{D}(x, -), H)$. Consider the functors $N, E : \mathbf{Set}^{\mathbf{D}} \times \mathbf{D} \rightarrow \mathbf{Set}$, the first defined as

$$\begin{array}{ccc} \mathbf{Set}^{\mathbf{D}} \times \mathbf{D} & & \mathbf{Set} \\ \begin{array}{c} H \\ \eta \Downarrow \\ G \end{array} & \begin{array}{c} x \\ \downarrow f \\ z \end{array} & \begin{array}{c} \text{Nat}(\mathbf{D}(x, -), H) \\ \xrightarrow{N} \downarrow (\eta, f)^N \\ \text{Nat}(\mathbf{D}(z, -), G) \end{array} \end{array}$$

where $(\eta, f)^N$ is the function that assigns to every natural transformation $\alpha : \mathbf{D}(x, -) \Rightarrow H$ the composite natural transformation $(f-)\alpha\eta$:

$$\begin{array}{ccc} \mathbf{D}(x, -) & \xrightarrow{\alpha} & H \\ \uparrow f- & & \downarrow \eta \\ \mathbf{D}(z, -) & & G \end{array}$$

whereas the other functor is described as:

$$\begin{array}{ccc}
 \mathbf{Set}^{\mathbf{D}} \times \mathbf{D} & & \mathbf{Set} \\
 \begin{array}{c} H \\ \eta \Downarrow \\ G \end{array} & \begin{array}{c} x \\ \downarrow f \\ z \end{array} & \xrightarrow{E} \begin{array}{c} x^H \\ \downarrow \eta_x f^G = f^H \eta_z \\ z^G \end{array}
 \end{array}$$

where the equality is justified by the fact that $(\eta, f)^E$ is precisely the diagonal of the square

$$\begin{array}{ccc}
 x^H & \xrightarrow{\eta_x} & x^G \\
 f^H \downarrow & & \downarrow f^G \\
 z^H & \xrightarrow{\eta_z} & z^G
 \end{array}$$

Theorem 2.1.19 (Yoneda Lemma). *Given a category $\mathbf{D}$, for every choice of a functor $H: \mathbf{D} \rightarrow \mathbf{Set}$ with $\mathbf{D}$ locally small and of an object x in $\mathbf{D}$, there exists a bijection $y_{H,x}$ from the class $\text{Nat}(\mathbf{D}(x, -), H)$ to the set x^H , from which it follows that the former is itself a set. The bijection is defined as:*

$$y_{H,x}: \alpha \in \text{Nat}(\mathbf{D}(x, -), H) \mapsto 1_x^{\alpha_x} \in x^H$$

Moreover, the bijections are natural in H and in x , that is, as the pair of objects (H, x) of $\mathbf{Set}^{\mathbf{D}} \times \mathbf{D}$ varies, the $y_{H,x}$ assemble into a natural isomorphism y :

$$\begin{array}{ccc}
 & N & \\
 \mathbf{Set}^{\mathbf{D}} \times \mathbf{D} & \begin{array}{c} \Downarrow y \\ \Downarrow y \end{array} & \mathbf{Set} \\
 & E &
 \end{array}$$

Proof. The proof of the bijectivity of $y_{H,x}$ proceeds similarly to the proof of part (i) of Proposition 2.1.12. Consider the function

$$T_{H,x}: u \in x^H \mapsto \hat{u} \in \text{Nat}(\mathbf{D}(x, -), H)$$

where $\hat{u} = (\hat{u}_d)_{d \in \text{obj} \mathbf{D}}$ with $\hat{u}_d: f \mapsto u^{f^H}$, which is shown to be a natural transformation analogously to Proposition 2.1.12. Given a natural transformation $\alpha: \mathbf{D}(x, -) \Rightarrow H$, that is, $\alpha \in \text{Nat}(\mathbf{D}(x, -), H)$, for every arrow $x \xrightarrow{f} d$ of $\mathbf{D}$, one has the commutative diagram

$$\begin{array}{ccc}
\mathbf{D}(x, x) & \xrightarrow{\alpha_x} & x^H \\
\downarrow -f & & \downarrow f^H \\
\mathbf{D}(x, d) & \xrightarrow{\alpha_d} & d^H
\end{array}$$

Focusing on how 1_x is transformed, one obtains:

$$f^{\alpha_d} = (1_x f)^{\alpha_d} = 1_x^{(-f)\alpha_d} = 1_x^{\alpha_x f^H} = (1_x^{\alpha_x})^{f^H}$$

that is, putting $u = 1_x^{\alpha_x}$, $\alpha_d: f \mapsto u^{f^H}$, namely $\alpha = \hat{u}$. Thus

$$\alpha^{y_{H,x} T_{H,x}} = u^{T_{H,x}} = \hat{u} = \alpha$$

whereas

$$u^{T_{H,x} y_{H,x}} = \hat{u}^{y_{H,x}} = 1_x^{\hat{u}_x} = u^{1_x^H} = u^{1_x^H} = u$$

therefore $y_{H,x}$ is invertible, hence a bijection.

It remains to prove that y is natural. Choose a natural transformation $\eta: F \Rightarrow G$ in $\mathbf{Set}^{\mathbf{D}}$ and an arrow $a \xrightarrow{f} b$ in $\mathbf{D}$; observe:

$$\begin{array}{ccc}
\text{Nat}(\mathbf{D}(a, -), F) & \xrightarrow{y_{F,a}} & a^F \\
\downarrow (\eta, f)^N & & \downarrow (\eta, f)^E \\
\text{Nat}(\mathbf{D}(b, -), G) & \xrightarrow{y_{G,b}} & b^G
\end{array}$$

if α is a natural transformation from $\mathbf{D}(a, -)$ to F , then:

$$\alpha^{(\eta, f)^N y_{G,b}} = ((f-) \alpha \eta)^{y_{G,b}} = 1_b^{((f-) \alpha \eta)_b} = f^{\alpha_b \eta_b}$$

by the naturality of α one obtains the following diagram, starting from the arrow $a \xrightarrow{f} b$:

$$\begin{array}{ccc}
\mathbf{D}(a, a) & \xrightarrow{\alpha_a} & a^F \\
\downarrow -f & & \downarrow f^F \\
\mathbf{D}(a, b) & \xrightarrow{\alpha_b} & b^F
\end{array}$$

therefore $1_a^{\alpha_a f^F} = 1_a^{(-f)\alpha_b} = f^{\alpha_b}$, which in the previous equation gives:

$$f^{\alpha_b \eta_b} = (1_a^{\alpha_a f^F})^{\eta_b} = (1_a^{\alpha_a})^{f^F \eta_b} = (1_a^{\alpha_a})^{(\eta, f)^E} = \alpha^{y_{F,a}(\eta, f)^E}$$

and the naturality of y follows. $\square$

With the notation above, the following functors, the second dual to the first, will be called *Yoneda functors*:

$$\begin{array}{ccc} \mathbf{D}^{\text{op}} & \mathbf{Set}^{\mathbf{D}} & \mathbf{D} \\ \begin{array}{c} x \\ f \downarrow \\ y \end{array} & \xrightarrow{Y} \begin{array}{c} \mathbf{D}(x, -) \\ \uparrow f- \\ \mathbf{D}(y, -) \end{array} & \begin{array}{c} x \\ f \downarrow \\ y \end{array} \xrightarrow{Y'} \begin{array}{c} \mathbf{D}(-, x) \\ \downarrow -f \\ \mathbf{D}(-, y) \end{array} \end{array}$$

These functors are often also called *Yoneda embeddings*: given two objects x and y of $\mathbf{D}$, a natural transformation $\alpha: \mathbf{D}(y, -) \Rightarrow \mathbf{D}(x, -)$ corresponds, by the Yoneda lemma, to an element $f := 1_y^{\alpha_y}$ of $\mathbf{D}(x, y)$; hence its components are $\alpha_d: g \in \mathbf{D}(y, d) \mapsto fg \in \mathbf{D}(x, d)$, that is, $\alpha_d = f-$ for every object d in $\mathbf{D}$. This argument and its dual prove the following important consequence of the Yoneda lemma.

Theorem 2.1.20. *The Yoneda functors are full and faithful.*

Example 2.1.21. Given a group G and the associated group category $\mathbf{G}$, applying the Yoneda functor to $\mathbf{G}$

$$\begin{array}{ccc} \mathbf{G}^{\text{op}} & \mathbf{Set}^{\mathbf{G}} & \\ \begin{array}{c} X \\ g \downarrow \\ X \end{array} & \xrightarrow{Y} \begin{array}{c} \mathbf{G}(X, -) \\ \uparrow g- \\ \mathbf{G}(X, -) \end{array} & \end{array}$$

one can observe that the functor $\mathbf{G}(X, -)$ can be applied only to the object X : the set $\mathbf{G}(X, X)$ constitutes the entire underlying set of the group G . Since functors send isomorphisms to isomorphisms and every element g of the group G is an automorphism of X , its image $g-$ under $\mathbf{G}(X, -)$ is also an automorphism of sets (a permutation) of $\mathbf{G}(X, X)$. In practice, the embedding describes a bijection ϕ between G and a set of permutations $\text{Cay}(G)$. The set of permutations S_T of a set T , under the usual composition of functions, is a group: by the functoriality of Y one finds that $\text{Cay}(G)$ is a subgroup of $S_{\mathbf{G}(X, X)}$ and that ϕ is in particular an isomorphism. What has just been observed is nothing other than a version of Cayley's theorem for groups: every group is isomorphic to a group of permutations of its underlying set.

Example 2.1.22. Returning to Example 2.1.15, for every vector space W of $\mathbf{Vect}_{\mathbb{K}}$ there exists a natural isomorphism, guaranteed by the Yoneda Lemma, of the form:

$$\text{Nat}(\mathbf{Vect}_{\mathbb{K}}(W, -), \text{Bil}_{U, V}) \cong W^{\text{Bil}_{U, V}}$$

In particular, if $W = U \otimes V$, then

$$\text{Nat}(\mathbf{Vect}_{\mathbb{K}}(U \otimes V, -), \text{Bil}_{U,V}) \cong (U \otimes V)^{\text{Bil}_{U,V}}$$

which suggests that if there is a natural isomorphism $\eta: \mathbf{Vect}_{\mathbb{K}}(U \otimes V, -) \Rightarrow \text{Bil}_{U,V}$, which is the definition of tensor product of vector spaces, then it is unique. That is, for every $\mathbb{K}$ -vector space X there exists a unique natural isomorphism

$$\eta_X: \mathbf{Vect}_{\mathbb{K}}(U \otimes V, X) \cong X^{\text{Bil}_{U,V}}$$

and this is precisely due to the existence of the universal arrow as described in Example 2.1.15.

Proposition 2.1.23. *Given two objects a and b of a locally small category $\mathbf{C}$, if one of the functors represented by a is naturally isomorphic to the functor of the appropriate variance represented by b , then a and b are isomorphic. Conversely, if a and b are isomorphic, then the functors, with the appropriate variance, represented by these two objects are naturally isomorphic.*

Proof. Let $\eta: \mathbf{C}(a, -) \Rightarrow \mathbf{C}(b, -)$ be a natural isomorphism. As a natural transformation, $\eta \in \text{Nat}(\mathbf{C}(a, -), \mathbf{C}(b, -))$, and by the Yoneda lemma $\text{Nat}(\mathbf{C}(a, -), \mathbf{C}(b, -)) \cong \mathbf{C}(b, a)$; hence η corresponds to a morphism $b \xrightarrow{f} a$ via $y_{\mathbf{C}(b,-),a}$. Since it is an isomorphism, η has an inverse $\eta^{-1}: \mathbf{C}(b, -) \Rightarrow \mathbf{C}(a, -)$, which, as before, corresponds to a morphism $a \xrightarrow{g} b$. Fixing the object a , use the naturality of $y_{-,a}$ on the arrow η^{-1} :

$$\begin{array}{ccc} \text{Nat}(\mathbf{C}(a, -), \mathbf{C}(b, -)) & \xrightarrow{y_{\mathbf{C}(b,-),a}} & \mathbf{C}(b, a) \\ \downarrow -\eta^{-1} & & \downarrow -g \\ \text{Nat}(\mathbf{C}(a, -), \mathbf{C}(a, -)) & \xrightarrow{y_{\mathbf{C}(a,-),a}} & \mathbf{C}(a, a) \end{array}$$

therefore:

$$(\text{id}_{\mathbf{C}(a,-)})^{y_{\mathbf{C}(a,-),a}} = (\eta\eta^{-1})^{y_{\mathbf{C}(a,-),a}} = \eta^{(-\eta^{-1})y_{\mathbf{C}(a,-),a}} = \eta^{y_{\mathbf{C}(b,-),a}(-g)} = f^{(-g)} = fg$$

but by definition $(\text{id}_{\mathbf{C}(a,-)})^{y_{\mathbf{C}(a,-),a}} = 1_a^{\text{id}_{\mathbf{C}(a,-)}} = 1_a$, hence $fg = 1_a$. Analogously one proves $gf = 1_b$, and therefore a and b are isomorphic. The second part of the statement is a direct consequence of Example 1.5.3. $\square$

It follows that a functor has at most one representing object up to isomorphism.

2.2 Adjunctions

Definition 2.2.1. Given two functors $F : \mathbf{C} \rightarrow \mathbf{D}$ and $G : \mathbf{D} \rightarrow \mathbf{C}$ between locally small categories, an **ADJUNCTION** from F to G is a natural isomorphism of the form

$$\begin{array}{ccc} & \mathbf{D}(-^F, -) & \\ \curvearrowright & \Downarrow \phi & \curvearrowleft \\ \mathbf{C}^{\text{op}} \times \mathbf{D} & & \mathbf{Set} \\ \curvearrowleft & \Downarrow & \curvearrowright \\ & \mathbf{C}(-, -^G) & \end{array}$$

One says that F is a *left adjoint* to G , and that G is a *right adjoint* to F . We shall use the notation $F \dashv G$ to indicate that there exists an adjunction making F a left adjoint to G .

Example 2.2.2. With the notation of Example 2.1.10, the following functors are adjoint:

$$\begin{array}{ccc} & L & \\ \curvearrowright & & \curvearrowleft \\ \mathbf{Set} & \perp & \mathbf{Vect}_{\mathbb{K}} \\ \curvearrowleft & & \curvearrowright \\ & U & \end{array}$$

Indeed, for every pair (S, V) consisting of a set and a vector space there is a natural isomorphism

$$\phi_{S,V}: S^L \xrightarrow{f} V \in \mathbf{Vect}_{\mathbb{K}}(S^L, V) \mapsto S \xrightarrow{f|_S} V^U \in \mathbf{Set}(S, V^U).$$

Example 2.2.3. Many examples of adjunctions involve forgetful functors U , such as:

$$\begin{array}{ccc} \begin{array}{ccc} & F & \\ \curvearrowright & & \curvearrowleft \\ \mathbf{Set} & \perp & \mathbf{Grp} \\ \curvearrowleft & & \curvearrowright \\ & U & \end{array} & \begin{array}{ccc} & F & \\ \curvearrowright & & \curvearrowleft \\ \mathbf{Set} & \perp & \mathbf{Ab} \\ \curvearrowleft & & \curvearrowright \\ & U & \end{array} & \begin{array}{ccc} & F & \\ \curvearrowright & & \curvearrowleft \\ \mathbf{Set} & \perp & \mathbf{Mod}_R \\ \curvearrowleft & & \curvearrowright \\ & U & \end{array} \end{array}$$

In the first case F assigns to each set its free group, and to each function $S \xrightarrow{f} T$ the homomorphism whose restriction to S agrees with f . In the other two cases F again provides free constructions, respectively free abelian groups and free R -modules; in particular, if $R = \mathbb{K}$ one recovers the previous example.

Let $\mathbf{Dom}_{\text{mon}}$ be the subcategory of $\mathbf{Ring}$ whose objects are integral domains and whose morphisms are only monic homomorphisms. Let $Q: \mathbf{Dom}_{\text{mon}} \rightarrow \mathbf{Field}$ be the functor assigning to each domain its field of fractions and to each monomorphism the map acting componentwise on numerator and denominator. Then the following adjunction holds, where U regards every field as an integral domain and every field morphism as a morphism of domains:

$$\begin{array}{ccc}
& Q & \\
\text{Dom}_{\mathbf{mon}} & \xrightarrow{\quad} & \mathbf{Field} \\
& U &
\end{array}$$

With suitable foundational choices, there would be a left adjoint F to the functor that assigns to every small category its underlying directed graph.

$$\begin{array}{ccc}
& F & \\
\mathbf{Graph} & \xrightarrow{\quad} & \mathbf{Cat} \\
& U &
\end{array}$$

The free category on an arbitrary graph is obtained by adjoining to the graph all identities and all formal composites.

The functor $-^\times: \mathbf{Ring} \rightarrow \mathbf{Grp}$ may be regarded as a forgetful functor and has as left adjoint the functor $\mathbb{Z}[-]: \mathbf{Grp} \rightarrow \mathbf{Ring}$, which forms a ring from formal sums of elements of G :

$$\begin{array}{ccc}
& \mathbb{Z}[-] & \\
\mathbf{Grp} & \xrightarrow{\quad} & \mathbf{Ring} \\
& U &
\end{array}$$

The forgetful functor from $\mathbf{Top}$ has both a left adjoint D and a right adjoint I : the former assigns to each set the topological space with that underlying set equipped with the discrete topology, while the latter equips it with the indiscrete topology.

$$\begin{array}{ccc}
& D & \\
\mathbf{Set} & \xleftarrow{\quad} & \mathbf{Top} \\
& I &
\end{array}$$

Indeed, for a pair (X, S) consisting of a topological space and a set, there is a natural isomorphism

$$\phi_{X,S}: X^U \xrightarrow{f} S \in \mathbf{Set}(X^U, S) \mapsto X \xrightarrow{f} S^I \in \mathbf{Top}(X, S^I),$$

since f is automatically continuous when S is given the indiscrete topology.

Example 2.2.4. View $\mathbb{R}$ and $\mathbb{Z}$ as poset categories with the usual order. There are adjunctions as described below:

$$\begin{array}{ccc}
& [-] & \\
\text{⤵} & \perp & \text{⤵} \\
\mathbb{R} & \xleftarrow{\quad} & \mathbb{Z} \\
\text{⤵} & \perp & \text{⤵} \\
& [-] &
\end{array}$$

Indeed, if (r, z) is a pair consisting of a real number and an integer, there is always a natural isomorphism $\mathbb{Z}(\lceil r \rceil, z) \cong \mathbb{R}(r, z)$: the two sets are empty if and only if $r > z$, while otherwise $\mathbb{Z}(\lceil r \rceil, z)$ contains the unique arrow $\lceil r \rceil \leq z$, which exists if and only if $r \leq z$. The other adjunction is treated analogously.

Example 2.2.5. A simple example of how universal quantifiers can be interpreted as adjunctions is the following. Let $p(x)$ be a proposition with a single free variable in some language $\mathcal{L}$. After fixing a model with underlying set X , one may assign to each $x \in X$ a truth value $P(x)$ of the formula $p(x)$: thus consider the function $X \xrightarrow{P} \Omega$, where $\Omega = \{\text{true}, \text{false}\}$ is a poset with order $\text{false} \leq \text{true}$. We also put a partial order on Ω^X by declaring, for $P, Q \in \Omega^X$ (assuming, though this is not necessary, that X and the language $\mathcal{L}$ are such that these functions arise from propositions), that $P \leq Q$ if and only if $P(x) \leq Q(x)$ for every $x \in X$. The set Ω^X has as its top and bottom elements, respectively, the constant functions **true** and **false**. If posets are viewed as poset categories, consider the functor $\forall_{x \in X} -: \Omega^X \rightarrow \Omega$, which sends $P \in \Omega^X$ to **true** if and only if $\forall x \in X P(x) = \text{true}$ (that is, if and only if it is the constant function **true**), and the functor $\exists_{x \in X} -: \Omega^X \rightarrow \Omega$, which sends $P \in \Omega^X$ to **true** if and only if $\exists x \in X P(x) = \text{true}$ (that is, it sends P to **false** if and only if P is the constant function **false**). In both cases morphisms are sent to the unique possible morphism in Ω . These functors satisfy the adjunctions described by

$$\begin{array}{ccc}
& \forall_{x \in X} & \\
\text{⤵} & \perp & \text{⤵} \\
\Omega^X & \xleftarrow{\quad} & \Omega \\
\text{⤵} & \perp & \text{⤵} \\
& \exists_{x \in X} &
\end{array}$$

where the inclusion functor is the one sending each truth value in Ω to the corresponding constant function, clearly preserving the order. Indeed, if one considers a pair (P, true) , there is an isomorphism

$$\Omega(\forall_{x \in X} P, \text{true}) \cong \Omega^X(P, \text{true})$$

simply because the order relations they describe are always satisfied; its naturality in the first variable is easy to check, as expressed by the following diagram.

$$\begin{array}{ccc}
\Omega(\forall_{x \in X} P, \text{true}) & \xrightarrow{\cong} & \Omega^X(P, \text{true}) \\
\leq - \uparrow & & \uparrow \leq - \\
\Omega(\forall_{x \in X} Q, \text{true}) & \xrightarrow{\cong} & \Omega^X(Q, \text{true})
\end{array}$$

A similar discussion applies to the other functor $\exists_{x \in X} -$.

Let $\phi_{-, -} : \mathbf{D}(-^F, -) \Rightarrow \mathbf{C}(-, -^G)$ be an adjunction from F to G . For every object c of $\mathbf{C}$, one obtains a representation

$$\phi_{c, -} : \mathbf{D}(c^F, -) \Rightarrow \mathbf{C}(c, -^G),$$

which, as seen in Proposition 2.1.12, corresponds to a universal arrow $\eta_c := 1_{c^F}^{\phi_{c, c^F}}$. As c varies among the objects of $\mathbf{C}$, the universal arrows η_c assemble into a natural transformation

$$\begin{array}{ccc}
& \text{id}_{\mathbf{C}} & \\
& \curvearrowright & \\
\mathbf{C} & \xrightarrow{\eta} & \mathbf{C} \\
& \curvearrowleft & \\
& FG &
\end{array}$$

Let us show the commutativity of the following diagram, for an arbitrary arrow $x \xrightarrow{f} y$, in order to prove the naturality of η :

$$\begin{array}{ccc}
x & \xrightarrow{\eta_x} & x^{FG} \\
\downarrow f & & \downarrow f^{FG} \\
y & \xrightarrow{\eta_y} & y^{FG}
\end{array}$$

By considering the two diagrams arising from the naturality of ϕ (whose components we shall not write explicitly for simplicity),

$$\begin{array}{ccc}
\mathbf{D}(x^F, x^F) & \xrightarrow{\phi} & \mathbf{C}(x, x^{FG}) \\
\downarrow -f^F & & \downarrow -f^{FG} \\
\mathbf{D}(x^F, y^F) & \xrightarrow{\phi} & \mathbf{C}(x, y^{FG})
\end{array}
\quad
\begin{array}{ccc}
\mathbf{D}(y^F, y^F) & \xrightarrow{\phi} & \mathbf{C}(y, y^{FG}) \\
\downarrow f^F - & & \downarrow f - \\
\mathbf{D}(x^F, y^F) & \xrightarrow{\phi} & \mathbf{C}(x, y^{FG})
\end{array}$$

one obtains the equality

$$\eta_x f^{FG} = \text{id}_{x^F} \phi f^{FG} = \text{id}_{x^F} \phi (-f^{FG}) = \text{id}_{x^F} (-f^F) \phi = \text{id}_{x^F} f^F \phi =$$

$$= f^F \text{id}_{y^F}^\phi = \text{id}_{y^F}^{(f^F -)^\phi} = \text{id}_{y^F}^{\phi(f-)} = f \text{id}_{y^F}^\phi = f \eta_y,$$

which proves the commutativity of the diagram and therefore the naturality of η .

Dually, given an object d of $\mathbf{D}$, each representation of $\mathbf{D}(-^F, d)$ corresponds to a universal arrow from F to d , denoted $\varepsilon_d := 1_{d^F}^\phi$. As the object d varies in $\mathbf{D}$, these transformations assemble into a natural transformation

$$\begin{array}{ccc} & GF & \\ \text{D} & \begin{array}{c} \xrightarrow{\quad} \\ \varepsilon \\ \xleftarrow{\quad} \end{array} & \text{D} \\ & \text{id}_{\text{D}} & \end{array}$$

The transformations η and ε are called, respectively, the *unit* and the *counit* of the adjunction, and they satisfy the following identities, called the *triangular identities*:

$$\begin{array}{ccc} G & \xrightarrow{G\eta} & GFG \\ & \searrow & \downarrow \varepsilon G \\ & & G \end{array} \quad \begin{array}{ccc} F & & \\ \eta F \downarrow & \searrow & \\ FGF & \xrightarrow{F\varepsilon} & F \end{array}$$

We prove the first one; the second is checked analogously. Given an arbitrary arrow $x \xrightarrow{f} y$ of $\mathbf{C}$, the naturality of η and ε on this arrow is expressed by the diagrams

$$\begin{array}{ccc} x & \xrightarrow{\eta_x} & x^{FG} \\ f \downarrow & & \downarrow f^{FG} \\ y & \xrightarrow{\eta_y} & y^{FG} \end{array} \quad \begin{array}{ccc} x^{GF} & \xrightarrow{\varepsilon_x} & x \\ f^{GF} \downarrow & & \downarrow f \\ y^{GF} & \xrightarrow{\varepsilon_y} & y \end{array}$$

whereas the naturality of the transformations $G\eta$ and εG gives the diagrams

$$\begin{array}{ccc} x^G & \xrightarrow{\eta_{x^G}} & x^{GFG} \\ f^G \downarrow & & \downarrow f^{GFG} \\ y^G & \xrightarrow{\eta_{y^G}} & y^{GFG} \end{array} \quad \begin{array}{ccc} x^{GFG} & \xrightarrow{\varepsilon_x^G} & x^G \\ f^{GFG} \downarrow & & \downarrow f^G \\ y^{GFG} & \xrightarrow{\varepsilon_y^G} & y^G \end{array}$$

which shows that indeed $(G\eta)(\varepsilon G) = \text{id}_G$.

Example 2.2.6. Returning to the discussion of the forgetful functors in Example 2.2.3, consider for instance the adjunction

$$\begin{array}{ccc}
& F & \\
\text{Set} & \begin{array}{c} \curvearrowright \\ \perp \\ \curvearrowleft \end{array} & \text{Mod}_R \\
& U &
\end{array}$$

Its unit is the natural transformation whose components are $S \xrightarrow{\eta_S} (S^F)^U$ for every set S , namely the function that injects each set into the underlying set of the free R -module it generates. Similarly, the counit has as components the arrows $(M^U)^F \xrightarrow{\varepsilon_M} M$ for every R -module M , sending each formal sum $\sum_i r_i \overline{m_i}$ to its evaluation in M , namely $\sum_i r_i m_i$.

Theorem 2.2.7. *An adjunction is uniquely determined by its unit and counit. Conversely, given two functors $F: \mathbf{C} \rightarrow \mathbf{D}$ and $G: \mathbf{D} \rightarrow \mathbf{C}$ between locally small categories, a pair of natural transformations $\eta: \text{id}_{\mathbf{C}} \Rightarrow FG$ and $\varepsilon: GF \Rightarrow \text{id}_{\mathbf{D}}$ satisfying the triangular identities uniquely determines an adjunction $F \dashv G$.*

Proof. The first part has already been proved. For the second, given an object c of $\mathbf{C}$ and an object d of $\mathbf{D}$, consider

$$\phi: \mathbf{D}(c^F, d) \rightarrow \mathbf{C}(c, d^G) \quad \text{with} \quad \phi: c^F \xrightarrow{f} d \mapsto c \xrightarrow{\eta_c} c^{FG} \xrightarrow{f^G} d^G$$

$$\psi: \mathbf{C}(c, d^G) \rightarrow \mathbf{D}(c^F, d) \quad \text{with} \quad \psi: c \xrightarrow{g} d^G \mapsto c^F \xrightarrow{g^F} d^{GF} \xrightarrow{\varepsilon_d} d.$$

These maps are inverses of each other, and therefore give the components of a natural isomorphism between the two functors $\mathbf{D}(-^F, d)$ and $\mathbf{C}(c, -^G)$. Indeed, for every arrow $c^F \xrightarrow{f} d$,

$$f^{\phi\psi} = (\eta_c f^G)^\psi = \eta_c^F f^{GF} \varepsilon_d.$$

By the naturality of ε one has $f^{GF} \varepsilon_d = \varepsilon_{c^F} f$, and hence

$$f^{\phi\psi} = \eta_c^F \varepsilon_{c^F} f.$$

Finally, observing that $(\eta F)_c = \eta_c^F$ and $(F\varepsilon)_c = \varepsilon_{c^F}$, we recover

$$f^{\phi\psi} = (\eta F)_c (F\varepsilon)_c f = (\eta F F \varepsilon)_c f = \text{id}_c f = f.$$

Thus ψ is a right inverse of ϕ ; dually, it is also a left inverse. □

Theorem 2.2.8. *An adjunction is completely determined by any of the following equivalent data:*

- (i) *two functors $F: \mathbf{C} \rightarrow \mathbf{D}$, $G: \mathbf{D} \rightarrow \mathbf{C}$ and a natural transformation $\eta: \text{id}_{\mathbf{C}} \Rightarrow FG$ such that for every object c of $\mathbf{C}$ the arrow $\eta_c: c \rightarrow c^{FG}$ is universal from c to G . In this case $F \dashv G$ is given by*

$$\phi: \mathbf{D}(-^F, -) \Rightarrow \mathbf{C}(-, -^G)$$

$$\phi_{c,d} : c^F \xrightarrow{f} d \mapsto c \xrightarrow{\eta_c} c^{FG} \xrightarrow{f^G} d^G.$$

- (ii) a functor $G : \mathbf{D} \rightarrow \mathbf{C}$ and, for every object c of $\mathbf{C}$, a universal arrow $c \xrightarrow{\eta_c} c_c^G$ from c to G . In this case there exists a left adjoint $F : \mathbf{C} \rightarrow \mathbf{D}$ to G , with $c^F = x_c$ for every object c in $\mathbf{C}$, and for every arrow $a \xrightarrow{f} b$ in $\mathbf{C}$ its image f^F is determined by the universal property of the arrows η_c :

$$\begin{array}{ccc} a & \xrightarrow{\eta_a} & a^{FG} \\ & \searrow f\eta_b & \downarrow f^{FG} \\ & & b^{FG} \end{array} \qquad \begin{array}{ccc} a^F & & \\ & \downarrow f^F & \\ & & b^F \end{array}$$

Thus η is a natural transformation, and this reduces to the previous item.

- (iii) two functors $F : \mathbf{C} \rightarrow \mathbf{D}$, $G : \mathbf{D} \rightarrow \mathbf{C}$ and a natural transformation $\varepsilon : GF \Rightarrow \text{id}_{\mathbf{D}}$ such that for every object d of $\mathbf{D}$ the arrow $\varepsilon_d : d^{GF} \rightarrow d$ is universal from F to d . In this case $F \dashv G$ is given by

$$\begin{aligned} \phi^{-1} : \mathbf{C}(-, -^G) &\Rightarrow \mathbf{D}(-^F, -) \\ \phi_{c,d}^{-1} : c &\xrightarrow{g} d^G \mapsto c^F \xrightarrow{g^F} d^{GF} \xrightarrow{\varepsilon_d} d. \end{aligned}$$

- (iv) a functor $F : \mathbf{C} \rightarrow \mathbf{D}$ and, for every object d of $\mathbf{D}$, a universal arrow $y_d^F \xrightarrow{\varepsilon_d} d$ from F to d . In this case there exists a right adjoint $G : \mathbf{D} \rightarrow \mathbf{C}$ to F , with $d^G = y_d$ for every object d in $\mathbf{D}$, and for every arrow $a \xrightarrow{f} b$ in $\mathbf{D}$ its image f^G is determined by the universal property of the arrows ε_d :

$$\begin{array}{ccc} b & \xleftarrow{\varepsilon_b} & b^{GF} \\ & \nwarrow \varepsilon_a f & \uparrow f^{GF} \\ & & a^{GF} \end{array} \qquad \begin{array}{ccc} b^G & & \\ & \uparrow f^G & \\ & & a^G \end{array}$$

Thus ε is a natural transformation, and this reduces to the previous item.

Also note that, if a functor $G : \mathbf{D} \rightarrow \mathbf{C}$ has two left adjoints $F, F' \dashv G$, then these are naturally isomorphic. Indeed, for every object c of $\mathbf{C}$, each adjunction determines a universal arrow $c \xrightarrow{\eta_c} c^{FG}$ and another $c \xrightarrow{\eta'_c} c^{F'G}$, both from c to G ; these are known to

be unique up to isomorphism. Hence there exists an isomorphism $\theta_x : c^F \rightarrow c^{F'}$, and it is easy to check that these assemble into a natural isomorphism $\theta : F \Rightarrow F'$.

We list a few notions before the next result. Let $F : \mathbf{C} \rightarrow \mathbf{D}$ be a functor. It is called *dense on objects* if, for every object d in $\mathbf{D}$, there exists an object c in $\mathbf{C}$ such that $c^F \cong d$. An *adjoint equivalence* between two categories $\mathbf{C}$ and $\mathbf{D}$ is an adjunction from $F : \mathbf{C} \rightarrow \mathbf{D}$ to $G : \mathbf{D} \rightarrow \mathbf{C}$ whose unit and counit are natural isomorphisms. We shall see that this condition is equivalent to an equivalence of categories; it is introduced here to emphasize that an equivalence is a stronger condition than an adjunction.

Theorem 2.2.9. *Given a functor $G : \mathbf{D} \rightarrow \mathbf{C}$, the following statements are equivalent:*

- (i) G is an equivalence of categories;
- (ii) G is full, faithful, and dense on objects;
- (iii) G is part of an adjoint equivalence.

Proof. (i) $\Rightarrow$ (ii): If G is an equivalence of categories, then there exists a functor $F : \mathbf{C} \rightarrow \mathbf{D}$ with two natural isomorphisms $\eta : \text{id}_{\mathbf{C}} \Rightarrow FG$ and $\varepsilon : GF \Rightarrow \text{id}_{\mathbf{D}}$. The functor G is dense because every object c of $\mathbf{C}$ is isomorphic to c^{FG} through η_c . For any two parallel arrows $f, g : x \rightarrow y$ in $\mathbf{D}$ such that $f^G = g^G$, one obtains $f^{GF} = g^{GF}$. Looking at the following diagram,

$$\begin{array}{ccccc}
 x & \xrightarrow{\varepsilon_x^{-1}} & x^{GF} & \xrightarrow{\varepsilon_x} & x \\
 \downarrow f & & \downarrow f^{GF} = g^{GF} & & \downarrow g \\
 y & \xleftarrow{\varepsilon_y} & y^{GF} & \xleftarrow{\varepsilon_y^{-1}} & y
 \end{array}$$

one obtains the equality $f = g$, and therefore G is faithful. Finally, first noting that the faithfulness of F is proved analogously, take any two objects x and y of $\mathbf{D}$ and any arrow $x^G \xrightarrow{f} y^G$ of $\mathbf{C}$. Define $g = \varepsilon_x^{-1} f^F \varepsilon_y$; from the following diagram one sees that $f^F = g^{GF}$:

$$\begin{array}{ccccc}
 x^{GF} & \xrightarrow{\varepsilon_x} & x & \xrightarrow{\varepsilon_x^{-1}} & x^{GF} \\
 \downarrow g^{GF} & & \downarrow g & & \downarrow f^F \\
 y^{GF} & \xleftarrow{\varepsilon_y^{-1}} & y & \xleftarrow{\varepsilon_y} & y^{GF}
 \end{array}$$

and from the faithfulness of F it follows that $g^G = f$, proving that G is full.

(ii) $\Rightarrow$ (iii): Suppose that G is full, faithful, and dense on objects. We must construct a left adjoint F to G whose unit and counit are isomorphisms. For every object c of $\mathbf{C}$, choose an object c^F of $\mathbf{D}$ such that $c \cong (c^F)^G$, which exists by density of G . Let η_c be an isomorphism from c to $(c^F)^G$. For every arrow $x \xrightarrow{f} y$ in $\mathbf{C}$, let f^F be the morphism (which exists by fullness and is unique by faithfulness) such that $(f^F)^G = f^{FG} = \eta_x^{-1} f \eta_y$:

$$\begin{array}{ccc} x & \xrightarrow{\eta_x} & x^{FG} \\ \downarrow f & & \downarrow f^{FG} \\ y & \xrightarrow{\eta_y} & y^{FG} \end{array}$$

From the diagram one also observes that the η_c assemble into a natural isomorphism $\eta : \text{id}_{\mathbf{C}} \Rightarrow FG$, which is the required unit. To define the counit, impose that it satisfy one of the two triangular identities (one easily verifies afterwards that it also satisfies the other), namely $(G\eta)(\varepsilon G) = \text{id}_G$. For every object c of $\mathbf{C}$ one must have

$$(G\eta)_c(\varepsilon G)_c = \eta_{c^G} \varepsilon_c^G = \text{id}_c,$$

so ε_c is recovered from the equation $\varepsilon_c^G = \eta_{c^G}^{-1}$ by the hypotheses on G . The arrow ε_c is an isomorphism because, by fullness, there exists an inverse g such that $\eta_{c^G} = g^G$. It is straightforward to verify that ε is natural and also satisfies the other triangular identity.

(iii) $\Rightarrow$ (i): If there is an adjoint equivalence between two categories $\mathbf{C}$ and $\mathbf{D}$, this means that there is an adjunction $F \vdash G$ between two functors $F : \mathbf{C} \rightarrow \mathbf{D}$ and $G : \mathbf{D} \rightarrow \mathbf{C}$ and that the unit and counit of the adjunction are natural isomorphisms; in particular $\text{id}_{\mathbf{C}} \cong FG$ and $GF \cong \text{id}_{\mathbf{D}}$, whence the assertion. $\square$

2.3 Limits

Definition 2.3.1. A **DIAGRAM** in a category $\mathbf{C}$ is a functor $D : \mathbf{J} \rightarrow \mathbf{C}$. In this context $\mathbf{J}$ is called the *index category* of D .

If the index category of a diagram is small, the diagram is called a *small diagram*. Given a diagram $D : \mathbf{J} \rightarrow \mathbf{C}$ in $\mathbf{C}$, for every object c of $\mathbf{C}$ let $\bar{c} : \mathbf{J} \rightarrow \mathbf{C}$ denote the diagram assigning to every object of $\mathbf{J}$ the object c and to every morphism of $\mathbf{J}$ the identity of c . Consider the *diagonal functor* $\Delta_D : \mathbf{C} \rightarrow \mathbf{C}^{\mathbf{J}}$ associated with D , described as follows:

$$\begin{array}{ccc}
\mathbf{C} & & \mathbf{C}^{\mathbf{J}} \\
\begin{array}{c} a \\ f \downarrow \\ b \end{array} & \xrightarrow{\Delta_D} & \begin{array}{c} \bar{a} \\ \Downarrow \bar{f} \\ \bar{b} \end{array}
\end{array}$$

where $\bar{f}$ is the natural transformation whose components are $\bar{f}_a = f$.

A natural transformation $v : \bar{c} \Rightarrow D$ is called a *cone* from c to D because, for every object i in $\mathbf{J}$, the components of v are of the form $i^{\bar{c}} \xrightarrow{v_i} i^D = c \xrightarrow{v_i} i^D$, that is, they all have c as their domain. The object c is called the *vertex* of the cone and D its *base*.

Definition 2.3.2. Given a diagram $D : \mathbf{J} \rightarrow \mathbf{C}$ of the category $\mathbf{C}$, a universal arrow (u, x) in $\mathbf{C}^{\mathbf{J}}$ from the functor Δ_D to the object D is a **LIMIT** of the diagram D , denoted by $D^{\text{Lim}}_{\leftarrow}$.

With the notation of the preceding definition, the cone from x to D is universal among cones to D :

$$\begin{array}{ccc}
D & \xleftarrow{u} & \bar{x} \\
& \swarrow v & \uparrow \bar{f} \\
& & \bar{c}
\end{array}
\qquad
\begin{array}{c}
x \\
\uparrow f \\
c
\end{array}$$

We shall often abuse terminology by referring to x as the limit of the diagram whenever the universal cone can be recovered from the context.

Dually, a natural transformation $v : D \Rightarrow \bar{c}$ is called a *cocone* from D to c because, for every object i in $\mathbf{J}$, the components of v are of the form $i^D \xrightarrow{v_i} i^{\bar{c}} = i^D \xrightarrow{v_i} c$, that is, they all have c as their codomain. As before, c is the *vertex* of the cocone and D its *base*.

Definition 2.3.3. Given a diagram $D : \mathbf{J} \rightarrow \mathbf{C}$ of the category $\mathbf{C}$, a universal arrow (u, x) in $\mathbf{C}^{\mathbf{J}}$ from the object D to the functor Δ_D is a **COLIMIT** of the diagram D , denoted by $D^{\text{Lim}}_{\rightarrow}$.

This information is summarized in the following diagram; equivalently, the cocone from D to x is universal among cocones from D :

$$\begin{array}{ccc}
D & \xrightarrow{u} & \bar{x} \\
& \searrow v & \downarrow \bar{f} \\
& & \bar{c}
\end{array}
\qquad
\begin{array}{c}
x \\
\downarrow f \\
c
\end{array}$$

Thus, essentially, colimits in $\mathbf{C}$ are limits in $\mathbf{C}^{\text{op}}$.

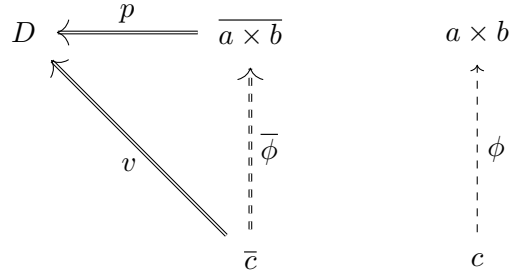
We list some of the most common index categories, and hence some classical types of (co)limits:

- (i) the category $\cdot \cdot$ has exactly two objects and only their identities as morphisms;
- (ii) given a class A , the category $\mathbf{points}_A$ is the discrete category whose objects are the elements of A , so $\cdot \cdot = \mathbf{points}_2$;
- (iii) the category $\cdot \rightarrow \cdot \leftarrow \cdot$ has three objects, which for simplicity we denote by α , β and ξ , and exactly two non-identity morphisms, one from α to ξ and one from β to ξ , so that objects and morphisms are arranged as in the displayed shape;
- (iv) the category $\cdot \rightrightarrows \cdot$ has two objects, denoted α and β , and exactly two distinct morphisms (besides the identities) from α to β .

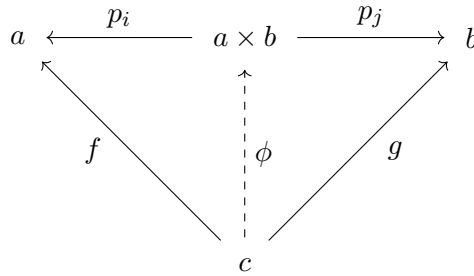
A *product* of two objects of $\mathbf{C}$ is the limit of a diagram D in $\mathbf{C}^{\cdot \cdot}$. If the two objects of the index category are i and j , with $i^D = a$ and $j^D = b$, the diagram may be represented as

$$a \quad b.$$

Then from $D \xleftarrow{\text{Lim}}$, which we call $a \times b$, there exists a cone p from $a \times b$ to the diagram which is universal with respect to every cone to D .



Since a cone v from an object c to the diagram consists only of a pair of arrows, say $c \xrightarrow{f} a$ and $c \xrightarrow{g} b$, we may say that, for any such pair of arrows, there exists a unique arrow $c \xrightarrow{\phi} a \times b$ making the following diagram commute:



Example 2.3.4. The product in $\mathbf{Set}$ is the usual Cartesian product, and its universal cone is given by the usual projections.

Example 2.3.5. The product in **Grp** is the direct product of groups, with the usual projections.

Example 2.3.6. The product in **Top** is the product of topological spaces equipped with the product topology. Indeed, one characterization of this space is precisely that the projections are continuous.

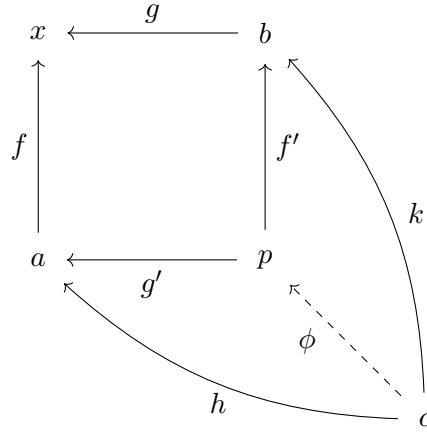
In general, given a class A of indices, a *product* of a family of objects of $\mathbf{C}$ indexed by A is the limit of a diagram D in $\mathbf{C}^{\mathbf{points}_A}$. If we set $c_\alpha := \alpha^D$ for every α in $\mathbf{points}_A$, we obtain the family of objects $(c_\alpha)_{\alpha \in A}$. Writing $\prod_{\alpha \in A} c_\alpha := D^{\text{Lim}}$, the existence of a universal cone p from $\prod_{\alpha \in A} c_\alpha$ to D is equivalent to saying that, for any family of arrows $(x \xrightarrow{f_\alpha} c_\alpha)_{\alpha \in A}$, with x an object of $\mathbf{C}$, there exists a unique arrow $x \xrightarrow{\phi} \prod_{\alpha \in A} c_\alpha$ making the following diagram commute for every $\alpha \in A$:

$$\begin{array}{ccc}
 c_\alpha & \xleftarrow{p_\alpha} & \prod_{\alpha \in A} c_\alpha \\
 & \nwarrow f_\alpha & \uparrow \phi \\
 & & x
 \end{array}$$

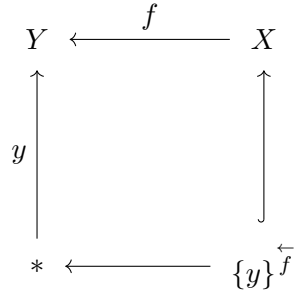
A *pullback* is the limit of a diagram D in $\mathbf{C}^{\rightarrow \leftarrow}$. If the shape $\alpha \rightarrow \xi \leftarrow \beta$ is sent by D to $a \xrightarrow{f} x \xleftarrow{g} b$, there exists a universal cone u from the pullback $p := D^{\text{Lim}}$ represented as follows:

$$\begin{array}{ccccc}
 a & \xrightarrow{f} & x & \xleftarrow{g} & b \\
 & \nwarrow u_a & \uparrow u_x & \nearrow u_b & \\
 & & p & &
 \end{array}$$

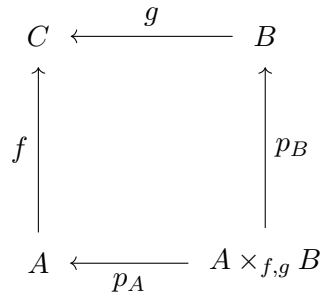
However, u_x is not information that is actually needed to construct the natural transformation u , because it is obtained from the composition $u_a f = u_b g$; an analogous observation holds for any cone to the diagram D . The morphism u_a is called the pullback of g along f , and u_b the pullback of f along g ; we shall denote u_a and u_b respectively by g' and f' . Therefore the universality of u is expressed by saying that, for every pair of arrows $a \xleftarrow{h} c \xrightarrow{k} b$, there exists a unique morphism $c \xrightarrow{\phi} p$ making the following diagram commute:



Example 2.3.7. In **Set**, given a function $X \xrightarrow{f} Y$ and an element $* \xrightarrow{y} Y$ of Y , a fiber is the pullback of the two arrows, as in the following diagram:



Given instead two functions $A \xrightarrow{f} C \xleftarrow{g} B$, there is their pullback as in the following diagram, identifying the projections with their restrictions:



where $A \times_{f,g} B$ is the set of pairs $(a, b) \in A \times B$ such that $a^f = b^g$. In **Set**, as in other concrete categories, saying that a pair of arrows f and g has a pullback amounts to saying that there is a set of solutions to the equation $x^f = x^g$ with $x \in A \times B$.

Example 2.3.8. On the multiplicative monoid $\mathbb{N}$ one may consider the divisibility relation $|$, which gives an order and hence allows us to view $\mathbb{N}$ as a poset category. Given two integers a and b , the evident relations $a|ab$ and $b|ab$ have as their pullback the greatest common divisor (a, b) :

$$\begin{array}{ccc}
ab & \longleftarrow & b \\
\uparrow & & \uparrow \\
a & \longleftarrow & (a, b)
\end{array}$$

An *equalizer* is the limit of a diagram D in $\mathbf{C}^{\rightrightarrows}$. If the shape $\alpha \rightrightarrows \beta$ is sent by D to $a \xrightarrow{f} b$, there exists a universal cone eq from the equalizer $e := D^{\text{Lim}}_{\leftarrow}$:

$$\begin{array}{ccc}
a & \xrightleftharpoons[f]{g} & b \\
\text{eq}_a \swarrow & & \searrow \text{eq}_b \\
& e &
\end{array}$$

However, since $\text{eq}_b = \text{eq}_a f = \text{eq}_a g$, the only information needed to construct the cone eq is the arrow eq_a ; accordingly, they may be identified. It is common to call the cone itself, namely the arrow eq , the equalizer, as well as its domain. Similarly, any cone to the diagram is determined by a single arrow $c \xrightarrow{h} a$, so for any such arrow there exists a unique $c \xrightarrow{\phi} e$ making the following diagram commute:

$$\begin{array}{ccccc}
e & \xrightarrow{\text{eq}} & a & \xrightleftharpoons[f]{g} & b \\
\uparrow \scriptstyle \phi & & \nearrow \scriptstyle h & & \\
c & & & &
\end{array}$$

Example 2.3.9. Observing that the pullback of two arrows with the same domain is in fact an equalizer, one is led to see that in **Set** an equalizer can be identified with the subset of the common domain on which the two arrows are equal:

$$E \hookrightarrow A \xrightleftharpoons[f]{g} B$$

Example 2.3.10. In **Grp**, the equalizer of a homomorphism $G \xrightarrow{f} H$ and the zero homomorphism $G \xrightarrow{n} H$ is the kernel of the homomorphism:

$$\ker(f) \hookrightarrow G \xrightleftharpoons[n]{f} H$$

By dualizing all the concepts mentioned above one obtains the corresponding colimits. First, however, observe that, since there are no morphisms besides identities, $\mathbf{points}_{\mathbf{A}}^{\text{op}} = \mathbf{points}_{\mathbf{A}}$; by sending one object to the other, one obtains an isomorphism of categories $\cdot \Leftarrow \cdot = (\cdot \Rightarrow \cdot)^{\text{op}} \cong \cdot \Rightarrow \cdot$; finally, $(\cdot \rightarrow \cdot \leftarrow \cdot)^{\text{op}} = \cdot \leftarrow \cdot \rightarrow \cdot$, defined analogously to the original one.

Given a class A of indices, the colimit of a diagram D in $\mathbf{C}^{\mathbf{points}_{\mathbf{A}}}$ is called the *coproduct* and is denoted by $\coprod_{\alpha \in A} c_{\alpha}$, where $\alpha^D = c_{\alpha}$ for every α in $\mathbf{points}_{\mathbf{A}}$. For every family of arrows $(c_{\alpha} \xrightarrow{f_{\alpha}} x)_{\alpha \in A}$ there exists a unique arrow $\coprod_{\alpha \in A} c_{\alpha} \xrightarrow{\phi} x$ making the following diagram commute for every α in A :

$$\begin{array}{ccc} c_{\alpha} & \xrightarrow{i_{\alpha}} & \coprod_{\alpha \in A} c_{\alpha} \\ & \searrow f_{\alpha} & \downarrow \phi \\ & & x \end{array}$$

Example 2.3.11. The coproduct in **Set** is the disjoint union of sets.

Example 2.3.12. The coproduct in **Grp** is the free product of groups.

Example 2.3.13. The coproduct in the category $\mathbf{Top}_{*}$, as in Example 1.4.3, is the wedge sum of topological spaces.

The colimit of a diagram D in $\mathbf{C}^{\leftarrow \rightarrow}$ is called a *pushout* p . If D sends $\alpha \leftarrow \xi \rightarrow \beta$ to $a \xleftarrow{f} x \xrightarrow{g} b$, then for every pair of arrows $a \xrightarrow{h} c \xleftarrow{k} b$ there exists a unique arrow $p \xrightarrow{\phi} c$ making the following diagram commute:

$$\begin{array}{ccccc} x & \xrightarrow{g} & b & & \\ \downarrow f & & \downarrow f' & \searrow k & \\ a & \xrightarrow{g'} & p & \xrightarrow{\phi} & c \\ & \searrow h & & & \end{array}$$

Example 2.3.14. Given two functions $A \xleftarrow{f} C \xrightarrow{g} B$, their pushout is as in the following diagram:

$$\begin{array}{ccc}
C & \xrightarrow{g} & B \\
f \downarrow & & \downarrow j_B \\
A & \xrightarrow{j_A} & A \sqcup_{f,g} B
\end{array}$$

where $A \sqcup_{f,g} B$ is the set $A \sqcup B$ quotiented by the relation $\sim$ defined as follows:

$$\forall x, y \in C \quad x \sim y \iff \exists z \in C \mid x = z^{f i_A} \wedge y = z^{g i_B},$$

while the functions j_A and j_B denote the canonical inclusions i_A and i_B composed with the projection onto the quotient set just described.

Example 2.3.15. View $\mathbb{N}$ as a poset category through the divisibility order. Given two integers a and b , the evident relations $1|a$ and $1|b$ have as their pushout the least common multiple $[a, b]$:

$$\begin{array}{ccc}
1 & \longrightarrow & b \\
\downarrow & & \downarrow \\
a & \longrightarrow & [a, b]
\end{array}$$

The colimit of a diagram D in $\mathbf{C}^{\rightrightarrows}$ is called the *coequalizer* e . If D sends $\alpha \rightrightarrows \beta$ to $a \xrightleftharpoons[f]{g} b$, then for every arrow $b \xrightarrow{h} c$ there exists a unique arrow $e \xrightarrow{\phi} c$ making the following diagram commute:

$$\begin{array}{ccccc}
a & \xrightleftharpoons[g]{f} & b & \xrightarrow{\text{coeq}} & e \\
& & \searrow h & & \downarrow \phi \\
& & & & c
\end{array}$$

Example 2.3.16. In $\mathbf{Grp}$, the coequalizer of a homomorphism $G \xrightarrow{f} H$ and the zero homomorphism $G \xrightarrow{n} H$ is the so-called cokernel of the homomorphism:

$$G \xrightleftharpoons[n]{f} H \xrightarrow{\pi_f} \text{coker}(f)$$

This construction is simply the quotient of the codomain group H by the normal subgroup generated by $\text{im}(f)$; in this way the identity element is identified with every element in the image of f .

Definition 2.3.17. A category is **COMPLETE** if it admits the limit of every small diagram. Dually, it is **COCOMPLETE** if it admits all such colimits. If it admits limits or colimits of every diagram indexed by a category with a finite number of objects, it is called *finitely complete* or *cocomplete*.

Fix a locally small category $\mathbf{C}$, an object c of it, and a small diagram D in it. We denote by $\text{Cone}(c, D) = \text{Nat}(\bar{c}, D)$ the set of all cones from c to D , while $\text{Cone}(D, c) = \text{Nat}(D, \bar{c})$ denotes the set of all cocones from D to c .

Theorem 2.3.18. **Set** is complete.

Proof. Let $D : \mathbf{J} \rightarrow \mathbf{Set}$ be a small diagram. We want to show that $D^{\text{Lim}} = (u, \text{Cone}(*, D))$, with $* = \{p\}$ the usual arbitrary singleton, and with cone $u : \overline{\text{Cone}(*, D)} \Rightarrow D$ whose components are the functions $u_i : \sigma \in \text{Cone}(*, D) \mapsto s_i \in i^D$, where, for every $i \in \text{obj} \mathbf{J}$, s_i denotes the image of the unique element of $*$ under the i -component σ_i of σ . If $i \xrightarrow{m} j$ is an arrow of $\mathbf{J}$, the commutativity of the following diagram

$$\begin{array}{ccc} i^D & \xrightarrow{m^D} & j^D \\ & \swarrow u_i & \nearrow u_j \\ & \text{Cone}(*, D) & \end{array}$$

follows from the naturality of each cone $\sigma \in \text{Cone}(*, D)$, which is expressed by the commutative diagram

$$\begin{array}{ccc} i^D & \xrightarrow{m^D} & j^D \\ & \swarrow \sigma_i & \nearrow \sigma_j \\ & * & \end{array}$$

from which one obtains

$$\sigma^{u_i m^D} = s_i^{m^D} = p^{\sigma_i m^D} = p^{\sigma_j} = s_j = \sigma^{u_j}.$$

Choose an arbitrary cone $\tau : \bar{X} \Rightarrow D$ from a set X to the diagram, and for every $x \in X$ denote by $* \xrightarrow{x_*} X$ the function sending the unique element of $*$ to x . To prove the universality of u , it remains to show that the function $\phi_\tau : x \in X \mapsto \overline{x_*} \tau \in \text{Cone}(*, D)$ is the unique function making the diagram commute:

$$\begin{array}{ccc}
D & \xleftarrow{u} & \overline{\text{Cone}(*, D)} \\
& \nwarrow \tau & \uparrow \overline{\phi_\tau} \\
& & \overline{X}
\end{array}$$

From the following diagrams

$$\begin{array}{ccc}
i^D & \xrightarrow{m^D} & j^D \\
\tau_i \swarrow & & \nearrow \tau_j \\
& X &
\end{array}
\quad
\begin{array}{ccc}
i^D & \xrightarrow{m^D} & j^D \\
x_*\tau_i \swarrow & & \nearrow x_*\tau_j \\
& * &
\end{array}$$

one obtains the well-definedness of ϕ_τ , namely that the functions $x_*\tau_i$ do indeed assemble into cones:

$$p^{x_*\tau_i m^D} = x^{\tau_i m^D} = x^{\tau_j} = p^{x_*\tau_j}.$$

Finally, for uniqueness, consider any arrow $X \xrightarrow{f} \text{Cone}(*, D)$ satisfying $f u_i = \tau_i$ for every object i in $\mathbf{J}$. For every $x \in X$, f determines a cone x^f , uniquely determined by its components $(x^f)_i$, which in turn are uniquely determined by the unique element in their image, namely

$$p^{(x^f)_i} = x^{f u_i} = p^{x_* f u_i} = p^{x_* \tau_i}.$$

Hence $x^f = x_* \tau = x^{\phi_\tau}$, which concludes the proof. $\square$

Theorem 2.3.19. *Given a category $\mathbf{J}$, if a category $\mathbf{C}$ admits equalizers of arbitrary pairs of parallel arrows and admits all products of families of objects indexed either by $\text{obj}\mathbf{J}$ or by $\text{mor}\mathbf{J}$, then $\mathbf{C}$ admits a limit for every diagram in $\mathbf{C}^{\mathbf{J}}$.*

Proof. Let D be such a diagram, and set $c_i := i^D$ for every object i in $\mathbf{J}$. The product of all these objects, namely the limit of the diagram in $\mathbf{C}^{\text{points}_{\text{obj}\mathbf{J}}}$ that assigns to every object i of $\mathbf{J}$ the corresponding c_i , will be denoted by $\prod_i c_i$, and its universal cone is denoted by

$$\begin{array}{ccc}
c_i & \xleftarrow{p_i} & \prod_i c_i \\
& \nwarrow f & \uparrow \phi \\
& & x
\end{array}$$

Also consider the product $\prod_{i \xrightarrow{u} j} c_j = \prod_u c_{\text{cod } u}$, namely the limit of the diagram in $\mathbf{Cpoints}_{\text{mor } \mathbf{J}}$ that assigns to every morphism $i \xrightarrow{u} j$ of $\mathbf{J}$ (which is now an object of $\mathbf{points}_{\text{mor } \mathbf{J}}$) the object c_j ; denote its cone by

$$\begin{array}{ccc} c_{\text{cod } u} & \xleftarrow{\pi_{\text{cod } u}} & \prod_u c_{\text{cod } u} \\ & \nwarrow f & \uparrow \phi \\ & & x \end{array}$$

Construct arrows f and g as in the figure from the universal property of $\prod_u c_{\text{cod } u}$, that is, let them be the two unique arrows making the following diagrams commute for every morphism $a \xrightarrow{v} b$ of $\mathbf{J}$:

$$\begin{array}{ccccc} c_{\text{cod } v} & \xleftarrow{\pi_{\text{cod } v}} & \prod_u c_{\text{cod } u} & & \prod_u c_{\text{cod } u} \xrightarrow{\pi_{\text{cod } v}} c_{\text{cod } v} \\ & \nwarrow v^D & \uparrow f & & \uparrow g \\ & c_{\text{dom } v} & \prod_i c_i & & \prod_i c_i \\ & \nwarrow p_{\text{dom } v} & & & \nearrow p_{\text{cod } v} \end{array}$$

Now, considering the equalizer $l \xrightarrow{\text{eq}} \prod_i c_i$ of f and g , define the arrow $\mu_a = \text{eq } p_a$ for every object c_a :

$$\begin{array}{ccc} l & \xrightarrow{\text{eq}} & \prod_i c_i \xrightleftharpoons[f]{g} \prod_u c_{\text{cod } u} \\ \mu_a \downarrow & \swarrow p_a & \\ c_a & & \end{array}$$

The arrows μ_i assemble into a cone μ from l to D :

$$\begin{array}{ccc} a^D & \xrightarrow{v^D} & b^D \\ \mu_a \swarrow & & \nearrow \mu_b \\ & l & \end{array}$$

Indeed, for every morphism $a \xrightarrow{v} b$ of $\mathbf{J}$, the commutativity of the preceding diagram is

shown by the chain of equalities

$$\mu_a v^D = \mu_{\text{dom } v} v^D = \text{eq } p_{\text{dom } v} v^D = \text{eq } f \pi_{\text{cod } v} = \text{eq } g \pi_{\text{cod } v} = \text{eq } p_{\text{cod } v} = \mu_{\text{cod } v} = \mu_b.$$

Choose an arbitrary cone τ from an object x of $\mathbf{C}$ to D . Construct h by the universal property of the product $\prod_i c_i$, namely as the unique arrow making the diagram commute for every c_i :

$$\begin{array}{ccc} c_i & \xleftarrow{p_i} & \prod_i c_i \\ & \nwarrow \tau_i & \uparrow h \\ & & x \end{array}$$

Moreover, there must exist a unique arrow making the following diagram commute:

$$\begin{array}{ccc} c_{\text{cod } u} & \xleftarrow{\pi_{\text{cod } u}} & \prod_u c_{\text{cod } u} \\ & \nwarrow \tau_{\text{cod } u} & \uparrow \\ & & x \end{array}$$

Since both hf and hg are suitable, they coincide; therefore h factors uniquely through the equalizer:

$$\begin{array}{ccccc} l & \xrightarrow{\text{eq}} & \prod_i c_i & \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} & \prod_u c_{\text{cod } u} \\ \uparrow k & & \nearrow h & & \\ x & & & & \end{array}$$

In turn, each component τ_i of τ factors through $\mu_i = \text{eq } p_i$:

$$\begin{array}{ccccc} l & \xrightarrow{\text{eq}} & \prod_i c_i & \xrightarrow{p_i} & c_i \\ \uparrow k & & \nearrow h & \searrow \tau_i & \\ x & & & & \end{array}$$

□

A direct consequence of this theorem is the following result, which applies, for instance, also to the case of **Set**.

Theorem 2.3.20. *If a category $\mathbf{C}$ admits every equalizer and every small product, then it is complete.*

Clearly all these results dualize to colimits and cocompleteness; **Set** is also cocomplete.

Chapter 3

Topos Theory

Definition 3.1. Let $\mathbf{C}$ be a locally small category with all binary products. We say that $\mathbf{C}$ has EXPONENTIAL OBJECTS if, for every object c , there exists a right adjoint $-^c$ to

$$\begin{array}{ccc} \mathbf{C} & & \mathbf{C} \\ a & \xrightarrow{- \times c} & a \times c \\ f \downarrow & & \downarrow f \times 1_c \\ b & & b \times c \end{array}$$

To examine this more explicitly, recall from item (iv) of Theorem 2.2.8 that an adjunction is completely determined by a functor and a universal arrow. Specifically, after fixing an object c and considering the functor $- \times c$, saying that the category $\mathbf{C}$ has exponentials is equivalent to saying that, for every object b of $\mathbf{C}$, there is a universal arrow $y_b \times c \xrightarrow{\text{ev}_b} b$ from $- \times c$ to b . In this way one constructs a right adjoint $-^c: \mathbf{C} \rightarrow \mathbf{C}$ by assigning to each object b in $\mathbf{C}$ the object $b^c := y_b$, and to each arrow $a \xrightarrow{f} b$ in $\mathbf{C}$ the arrow uniquely determined by the universal arrow $b^c \times c \xrightarrow{\text{ev}_b} b$ as follows:

$$\begin{array}{ccc} b & \xleftarrow{\text{ev}_b} & b^c \times c \\ & \nwarrow \text{ev}_a f & \uparrow f^c \times 1_c \\ & & a^c \times c \end{array} \qquad \begin{array}{c} b^c \\ \uparrow f^c \\ a^c \end{array}$$

Also observe that the adjunction yields a natural isomorphism $\mathbf{C}(- \times c, -) \cong \mathbf{C}(-, -^c)$, that is, to each arrow $a \times c \xrightarrow{f} b$ there corresponds uniquely the arrow ϕ determined by the universality of ev_b :

$$\begin{array}{ccccc}
b & \xleftarrow{\text{ev}_b} & b^c \times c & & b^c \\
& \nwarrow f & \uparrow \phi \times 1_c & & \uparrow \phi \\
& & a \times c & & a
\end{array}$$

Example 3.2. In the category **Set** exponentiation exists and coincides with the usual one. If one considers a function $A \times C \xrightarrow{f} B$, which one would say has two arguments, then, as above, f corresponds to a unique function $A \xrightarrow{\phi} B^C$. The function ϕ assigns to each element a of A the usual partial function $C \xrightarrow{f(a,-) := f_a} B$, namely the one obtained by keeping the first argument constant at a .

Definition 3.3. A category is called **CARTESIAN CLOSED** if it is finitely complete and has exponentials.

In particular, a finitely complete category has a terminal object because this is the limit of a diagram indexed by the empty category, which we may denote by **points**_∅: a terminal object is the product of an empty family of objects, that is, a product with no projections.

Definition 3.4. Let **C** be a category with a terminal object 1. A **SUBJECT CLASSIFIER** is a morphism $1 \xrightarrow{\text{true}} \Omega$ such that, for every monomorphism $c \xrightarrow{f} x$ in **C**, there exists a unique arrow $x \xrightarrow{\chi} \Omega$ making c a pullback as represented below:

$$\begin{array}{ccc}
c & \xrightarrow{f} & x \\
\downarrow & & \downarrow \chi \\
1 & \xrightarrow{\text{true}} & \Omega
\end{array}$$

Note that any arrow with domain 1 is a monomorphism: given $1 \xrightarrow{f} a$, any two parallel arrows $g, h : b \rightarrow 1$ such that $gf = hf$ are equal a priori, because there is a unique arrow from b to 1. In particular, the classifier is always a monomorphism. Moreover, for any object c of a category **C** with terminal object, the unique arrow from 1 to the object c will be denoted, when useful, by t_c .

Definition 3.5. In a category **C**, a **SUBJECT** of an object x is an equivalence class of a monomorphism in **C**/**x** modulo the relation of isomorphism.

Notice that, a priori, this equivalence class is not necessarily a set, and that all elements of the class are necessarily monomorphisms. In practice one often improperly refers to monomorphisms into x themselves as subobjects of x . The class of all subobjects of x will be denoted by $\text{sub}(x)$. A category in which all classes of subobjects are sets is called

well-powered. Thus, for a well-powered category $\mathbf{C}$ with pullbacks, one may consider the functor $\text{sub}: \mathbf{C}^{\text{op}} \rightarrow \mathbf{Set}$ which assigns to each object x its set $\text{sub}(x)$ and to each arrow $x \xrightarrow{f} y$ the arrow $\text{sub}(y) \xrightarrow{f^*} \text{sub}(x)$ mapping each class $[a \xrightarrow{m} y]$ to the class $[a' \xrightarrow{m'} x]$ whose representative is a pullback of $a \xrightarrow{m} y$ along f :

$$\begin{array}{ccc} a' & \longrightarrow & a \\ \downarrow m' & & \downarrow m \\ x & \xrightarrow{f} & y \end{array}$$

Indeed, the pullback of any monomorphism is a monomorphism. Given the following pullback square, draw two arrows x and y such that $xf = yf$. Then $xfi = yfi$, that is, $xi'm = yi'm$, and therefore $xi' = yi'$. Given the pair of arrows $xi' = yi'$ and $xf = yf$, there exists a unique arrow from d to p making the diagram commute, hence $x = y$.

$$\begin{array}{ccc} d & & \\ \downarrow x \quad \downarrow y & & \\ p & \xrightarrow{i'} & b \\ \downarrow f & & \downarrow m \\ a & \xrightarrow{i} & c \end{array}$$

Given these definitions, the classifier may be characterized as a universal element (dually, because of the contravariance of the functor) of sub :

$$\begin{array}{ccc} * & \xrightarrow{[\text{true}]} & \text{sub}(\Omega) \\ & \searrow [f] & \downarrow \chi^* \\ & & \text{sub}(x) \end{array} \quad \begin{array}{ccc} c & \xrightarrow{f} & x \\ \downarrow & & \downarrow \chi \\ 1 & \xrightarrow{\text{true}} & \Omega \end{array}$$

The original definition of elementary topos, as given by Lawvere and Tierney in 1969 in order to make it less restrictive than what is now called a *Grothendieck topos*, required an elementary topos to be a finitely complete and finitely cocomplete category with exponentials and a subobject classifier. Equivalently, one may say that it had to be cartesian closed (finitely complete with exponentials), finitely cocomplete, and equipped with a classifier. However, Mikkelsen later discovered that a cartesian closed category with a subobject classifier is always cocomplete, giving the following definition.

Definition 3.6. An ELEMENTARY TOPOS is a cartesian closed category with a subobject classifier.

We shall often refer to these simply as topoi. The adjective *elementary* is due to the fact that, while Grothendieck gave a definition of topos strongly tied to geometry, Lawvere and Tierney gave an intrinsically categorical one, more precisely one that could be formulated in the *Elementary* Theory of an Abstract Category (ETAC).

Example 3.7. The most classical example of a topos is the category **Set**. It is well known that **Set** is cartesian closed and has a subobject classifier, namely the arrow $*$ $\xrightarrow{\text{true}}$ 2 assigning the value 1 to the unique element of $*$. Given a set A and a subset $X \hookrightarrow A$, one obtains the diagram

$$\begin{array}{ccc} X & \xhookrightarrow{f} & A \\ \downarrow & & \downarrow \chi \\ * & \xrightarrow{\text{true}} & 2 \end{array}$$

where χ is the characteristic function of X in A , and is the unique function making the preceding square a pullback.

Proposition 3.8. *Given a topos $\mathbf{C}$, if an arrow $a \xrightarrow{f} b$ is both monic and epic, then it is an isomorphism.*

Proof. Using the universal property of the classifier on f and also drawing the terminal arrow of b , observe:

$$\begin{array}{ccc} a & \xrightarrow{f} & b \\ t_a \downarrow & \swarrow t_b & \downarrow \chi \\ 1 & \xrightarrow{\text{true}} & \Omega \end{array}$$

Note that $t_a = ft_b$ because 1 is terminal; moreover, writing $\text{true}_b := t_b \text{true}$, one has $f \text{true}_b = t_a \text{true}$. We want to show that f is an equalizer of χ and true_b . First observe that, since f is epic, $\chi = \text{true}_b$. Thus, given any arrow $c \xrightarrow{g} b$, all of this can be rewritten as a pullback diagram:

$$\begin{array}{ccccc} c & & \xrightarrow{g} & & b \\ & \searrow \text{dashed} & & \searrow f & \\ & & a & \xrightarrow{f} & b \\ t_c \downarrow & & t_a \downarrow & \swarrow t_b & \downarrow \chi \\ & & 1 & \xrightarrow{\text{true}} & \Omega \end{array}$$

that is, g factors uniquely through f ; hence f is an equalizer of χ and true_b . Since 1_b equalizes the two arrows, one obtains the following diagram:

$$\begin{array}{ccc} a & \xrightarrow{f} & b \xrightarrow[\text{true}_b]{\chi} \Omega \\ \uparrow \phi & \nearrow 1_b & \\ b & & \end{array}$$

It follows immediately that $\phi f = 1_b$, while

$$f\phi f = f1_b = f = 1_a f,$$

and since f is monic, $f\phi = 1_a$. Therefore f is an isomorphism. $\square$

Topoi have many properties which, like this one, make them resemble the category **Set**. For instance, one can show that every morphism $a \xrightarrow{f} b$ of a topos factors as $a \xrightarrow{f} b = a \xrightarrow{e} x \xrightarrow{m} b$, that is, as an epimorphism followed by a monomorphism. In **Set**, for example, this factorization is obtained by taking x to be the quotient of a by the relation induced by f .

Some properties studied in Topos Theory are the following, and they make it possible to distinguish topoi that are more or less close to the topos **Set**:

- A topos is called *well-pointed* if it has no zero objects and if, for every pair of parallel arrows $f, g: a \rightarrow b$, the condition $xf = xg$ for every arrow $1 \xrightarrow{x} a$ implies $f = g$.
- Given a topos **C**, a *natural numbers object* (NNO) is a triple $(N, 0, s)$ where N is an object and $1 \xrightarrow{0} N$ and $N \xrightarrow{s} N$ are morphisms such that, for every triple (X, z, r) , there exists a unique arrow $N \xrightarrow{f} X$ making the following diagram commute:

$$\begin{array}{ccccc} & & N & \xrightarrow{s} & N \\ & \nearrow 0 & \vdots & & \vdots \\ 1 & & f & & f \\ & \searrow z & \downarrow & & \downarrow \\ & & X & \xrightarrow{r} & X \end{array}$$

- A topos is said to satisfy *choice* if, for every epimorphism $a \xrightarrow{e} b$, there always exists a morphism $b \xrightarrow{f} a$ such that $fe = 1_b$.

Indeed, the category **Set** is well-pointed, since the empty set is not a singleton and every function is uniquely determined by how it acts on each element of its domain. The NNO is $(\mathbb{N}_0, 0, s)$, where 0 is the element $0 \in \mathbb{N}_0$ and s is the successor function. The condition is satisfied because it essentially summarizes the Induction Theorem, and the triple is what one calls a *Peano triple*. Saying that every surjection $a \xrightarrow{e} b$ has a section

means that there exists a function $b \xrightarrow{f} a$ such that $\forall x \in b \ x^f \in \{x\}^{\leftarrow e}$, by the axiom of choice.

Example 3.9. Expanding on what was seen in Example 1.4.10, a topological *sheaf* is a presheaf $F: \text{open}(X)^{\text{op}} \rightarrow \mathbf{Set}$, with X a topological space, satisfying the *gluing property*: if $U = \bigcup_{i \in J} U_i$ with $(U_i)_{i \in J}$ a family of open subsets of X , then for every family $(s_i)_{i \in J}$ of sections such that $s_i \in U_i^F \ \forall i \in J$, there exists a unique section $s \in U^F$ such that $s^{\text{res}_{U, U_i}} = s_i$. The gluing property can clearly also be generalized to more general sheaves not necessarily of a topological nature.

Let $\mathbf{Sh}(\mathbf{C})$ denote the category of sheaves (also non-topological ones) on the category $\mathbf{C}$, with natural transformations between them as morphisms, and let $\mathbf{PSh}(\mathbf{C})$ denote the analogous category of presheaves on $\mathbf{C}$. It can be shown that, if $\mathbf{C}$ is small, then $\mathbf{PSh}(\mathbf{C})$ is a topos and, in particular, so is $\mathbf{Sh}(\mathbf{C})$. Thus $\mathbf{Sh}(X) := \mathbf{Sh}(\text{open}(X))$ is a topos.

Let us describe its subobject classifier. The terminal object is the functor 1 sending every open set U to a set consisting of a section $*_U$ over U that is constantly equal to a point $*$ of X . Let Ω be the functor sending every open set U to the set $\{\text{id}_V \mid V \in \text{open}(U)\}$. The classifier $\text{true}: 1 \Rightarrow \Omega$ has components $\text{true}_U: *_U \in U^1 \mapsto \text{id}_U \in U^\Omega$. It will be convenient to identify U^Ω with $\text{open}(U)$. Given a monomorphism $m: G \Rightarrow F$, one can show that χ is the unique arrow making G a pullback as described by the following diagram:

$$\begin{array}{ccc} G & \xrightarrow{m} & F \\ \downarrow & & \vdots \chi \\ 1 & \xrightarrow{\text{true}} & \Omega \end{array}$$

in which χ has components χ_U assigning to each section $s \in U^F$ its largest open subset $V \in \text{open}(U) = U^\Omega$ such that $s|_V \in U^G$.

This topos is not well-pointed. Consider the sheaf ch_V that sends every open set U to $\{*_U\}$ if and only if $U \subseteq V$, and otherwise to the empty set. Take two open sets $N, M \in \text{open}(V)$ and construct χ^N and χ^M as follows:

$$\begin{array}{ccc} \text{ch}_N & \xrightarrow{\quad} & \text{ch}_V \\ \downarrow & & \vdots \chi^N \\ 1 & \xrightarrow{\text{true}} & \Omega \end{array} \quad \begin{array}{ccc} \text{ch}_M & \xrightarrow{\quad} & \text{ch}_N \\ \downarrow & & \vdots \chi^M \\ 1 & \xrightarrow{\text{true}} & \Omega \end{array}$$

There are no transformations $x: 1 \Rightarrow \text{ch}_V$, because their components would have to make the following diagram commute for every pair of open sets $A \subseteq B$:

$$\begin{array}{ccc}
\{*_A\} & \xrightarrow{x_A} & A^{\text{ch}_V} \\
\text{res} \downarrow & & \downarrow \text{res} \\
\{*_B\} & \xrightarrow{x_B} & A^{\text{ch}_V}
\end{array}$$

but, if at least A were empty, x_A would be a function with empty codomain. Therefore the condition $x\chi^N = x\chi^M$ for every $x: 1 \Rightarrow \text{ch}_V$ is vacuously satisfied. However, the component χ_V^N of χ^N assigns to the unique section $*_V \in V^{\text{ch}_V}$ the largest open set U such that the restriction $*_V|_U = *_U$ lies in U^{ch_N} , namely $U = N$. Analogously, χ_V^M sends $*_V$ to M , so χ^M and χ^N are distinct; hence the topos is not well-pointed.

Example 3.10. The category **FinSet** is the full subcategory of **Set** whose objects are only the finite sets. It is clearly also a topos, for the same reasons that **Set** is. This topos, however, has no NNO. Indeed, if $(N, 0, s)$ were an NNO for **FinSet**, one could show that $0 \notin \text{im } s$ and that s is injective: this is quite reasonable, since these are the axioms that a Peano triple must satisfy. On finite sets, injectivity is equivalent to bijectivity, so $\text{im } s$ has as many elements as N and is a subset of it; hence $\text{im } s = N$, but then $0 \in \text{im } s$, a contradiction. Saying that a topos has the induction principle also entails that it has an object with an *infinite number of elements*; that is, the presence of an NNO also expresses the axiom of infinity, and it is reasonable that **FinSet** cannot have one.

Example 3.11. Consider the category $\mathbf{G}\text{-Set} = \mathbf{Set}^{\mathbf{G}}$. Each of its object-functors sends the unique object of the category $\mathbf{G}$ to a specific set and all its morphisms to permutations of that set, respecting functoriality; these functors therefore correspond to actions of a group on a given set, and we shall call them *G-sets*.

The category $\mathbf{G}$ is isomorphic to $\mathbf{G}^{\text{op}}$ through the functor $-^{-1}$. It is then easy to see that the precomposition functor by $-^{-1}$, which we call $P: \mathbf{G}\text{-Set} \rightarrow \mathbf{PSh}(\mathbf{G})$, makes the two categories isomorphic. In particular $\mathbf{G}\text{-Set}$ is a topos, but it does not satisfy choice. For example, let $G = \mathbb{Z}_2$ be additive. Consider the nontrivial action on its underlying set $S := \{0, 1\}$, which consists of only two permutations: $0 + (-): S \rightarrow S$, the identity, and $1 + (-): S \rightarrow S$, which swaps the two elements. On the terminal object $*$, instead, there is only one possible action, namely the trivial one. In **Set**, consider the terminal map $S \xrightarrow{t} *$, which is equivariant with respect to the only possible actions and therefore induces a morphism t^* between the $\mathbb{Z}_2$ -sets. Fix on S the action $1 + (-)$, and on $*$ the action represented by 1. The map t^* is an epimorphism because, given two equivariant maps $* \xrightarrow{f} A$ and $* \xrightarrow{g} A$, if $t^*f^* = t^*g^*$, then in particular, since there is only one component, $tf = tg$; and since f and g have only one element in their image, as does t , it follows that $f = g$. Hence t^* is an epimorphism. If $\mathbf{G}\text{-Set}$ satisfied choice, then t^* would have a section s^* , that is, $st = 1_*$ with $* \xrightarrow{s} S$. Then, by the equivariance of s , one would have

$$1 + p^s = (1 + p)^s,$$

that is, $1 = 0$, whatever the element in the image of s is.

In 1964 Lawvere put forward a radical proposal: to take the above properties as axioms for sets. The *Elementary Theory of the Category of Sets* (ETCS) is an elementary theory that axiomatizes the category of sets as a well-pointed topos with an NNO and satisfying choice. In principle, it could be an effective alternative to classical ZFC set theory; however, it does not include an analogue of the axiom schema of replacement.

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